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(54) **DEVELOPING APPARATUS AND IMAGE FORMING APPARATUS**

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(57) **ABSTRACT**

A developing apparatus comprises a development chamber with a developer bearing member and supply member, and a developer accommodation chamber above it. A stirring member with a sheet member rotates to stir developer. All members rotate along the longitudinal direction, with an opening along the chamber walls. In cross-section, the opening has upstream and downstream ends relative to the stirring member's rotation. At the longitudinal center, the distance from the upstream end to the contact region between the developer bearing member and supply member is shorter than the same distance measured at the longitudinal end.

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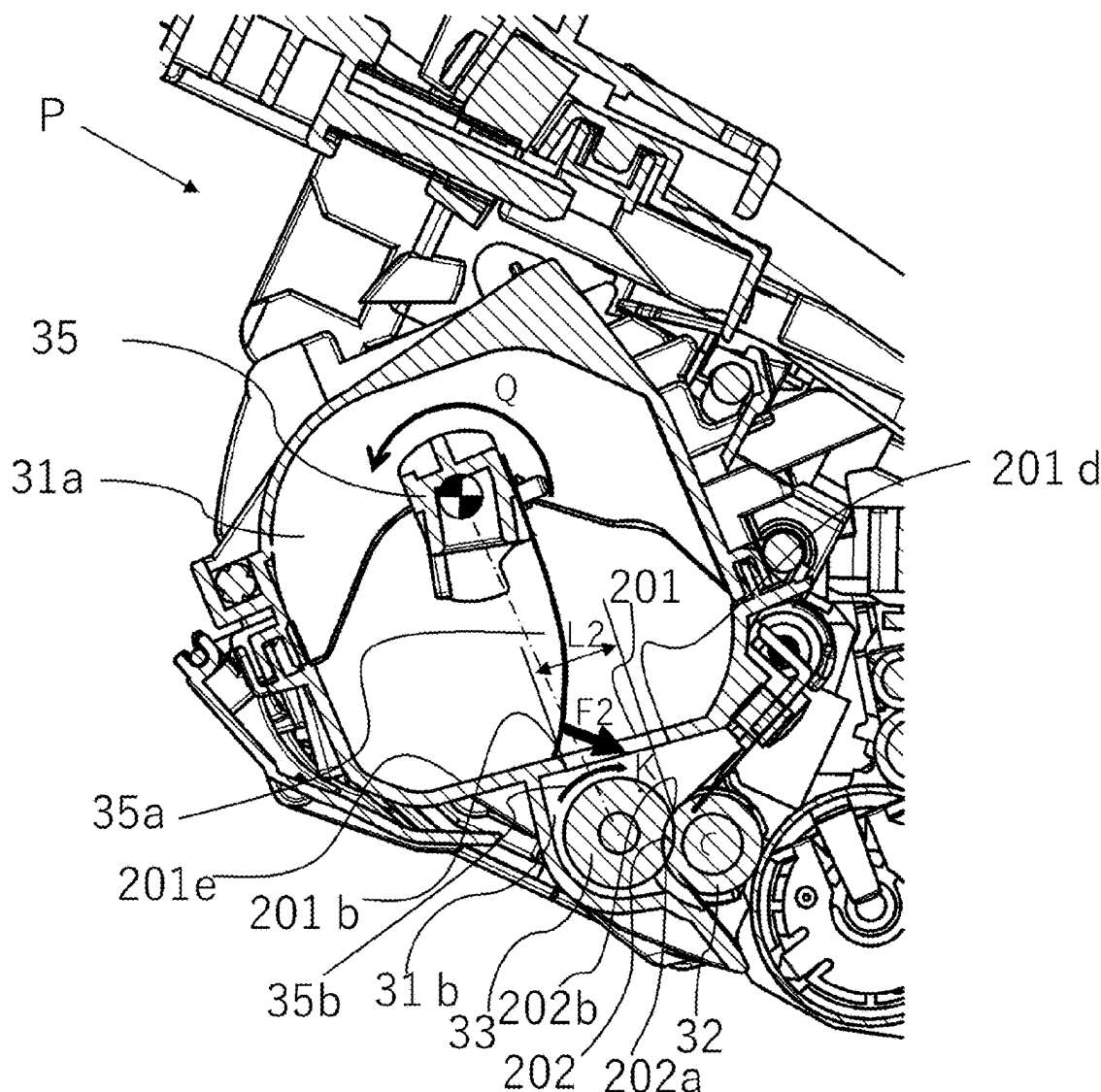


FIG. 1B

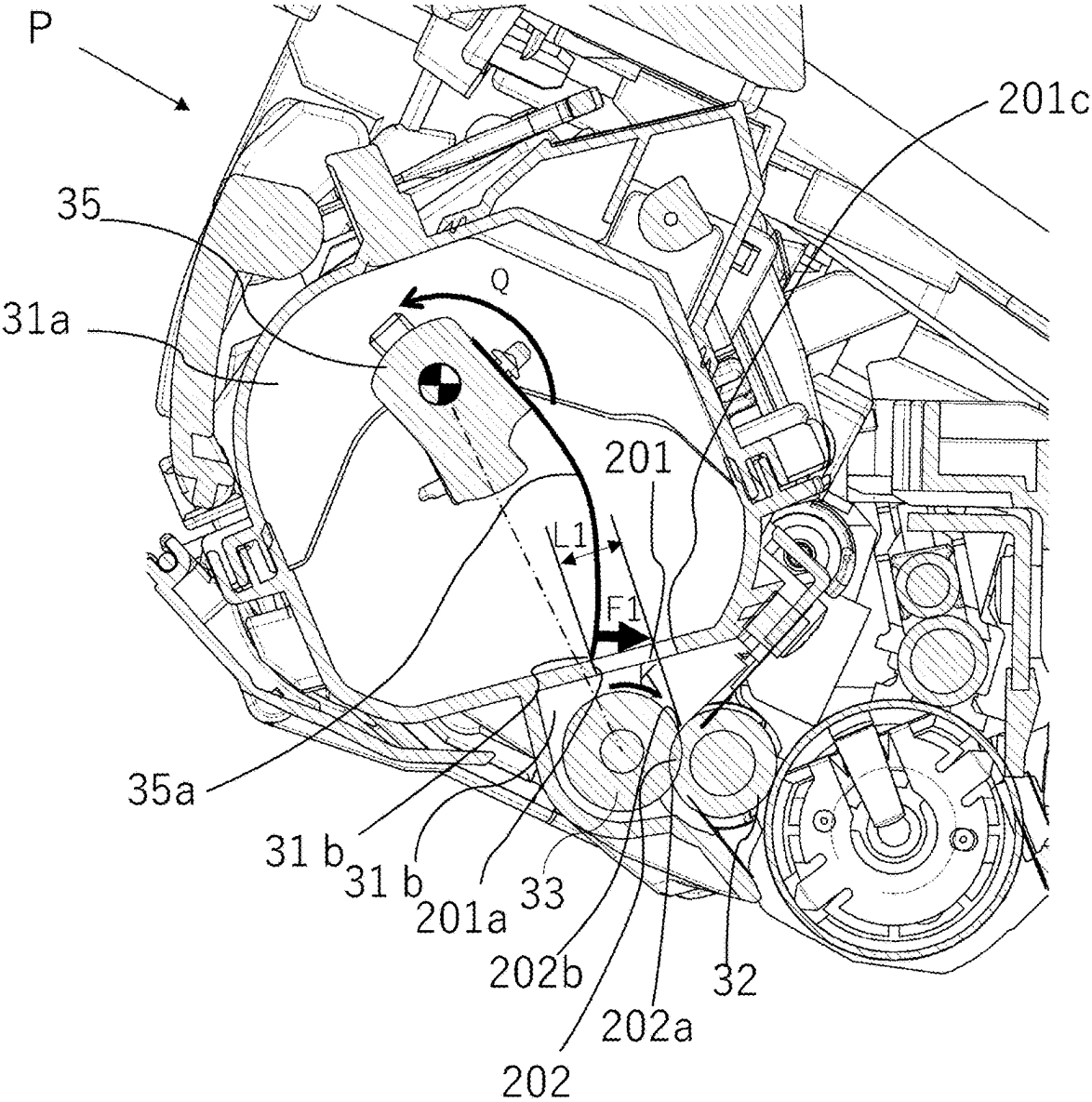
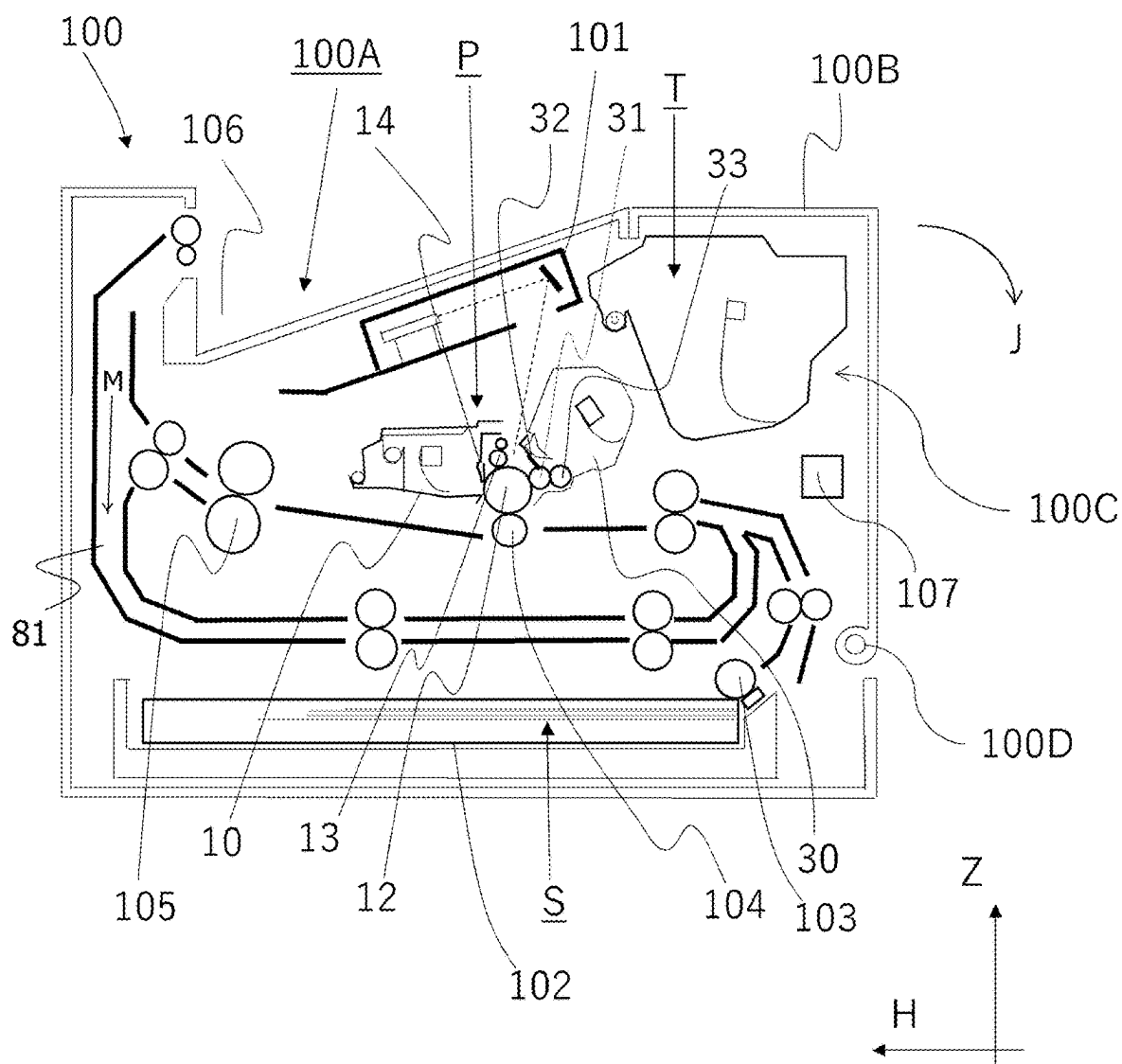
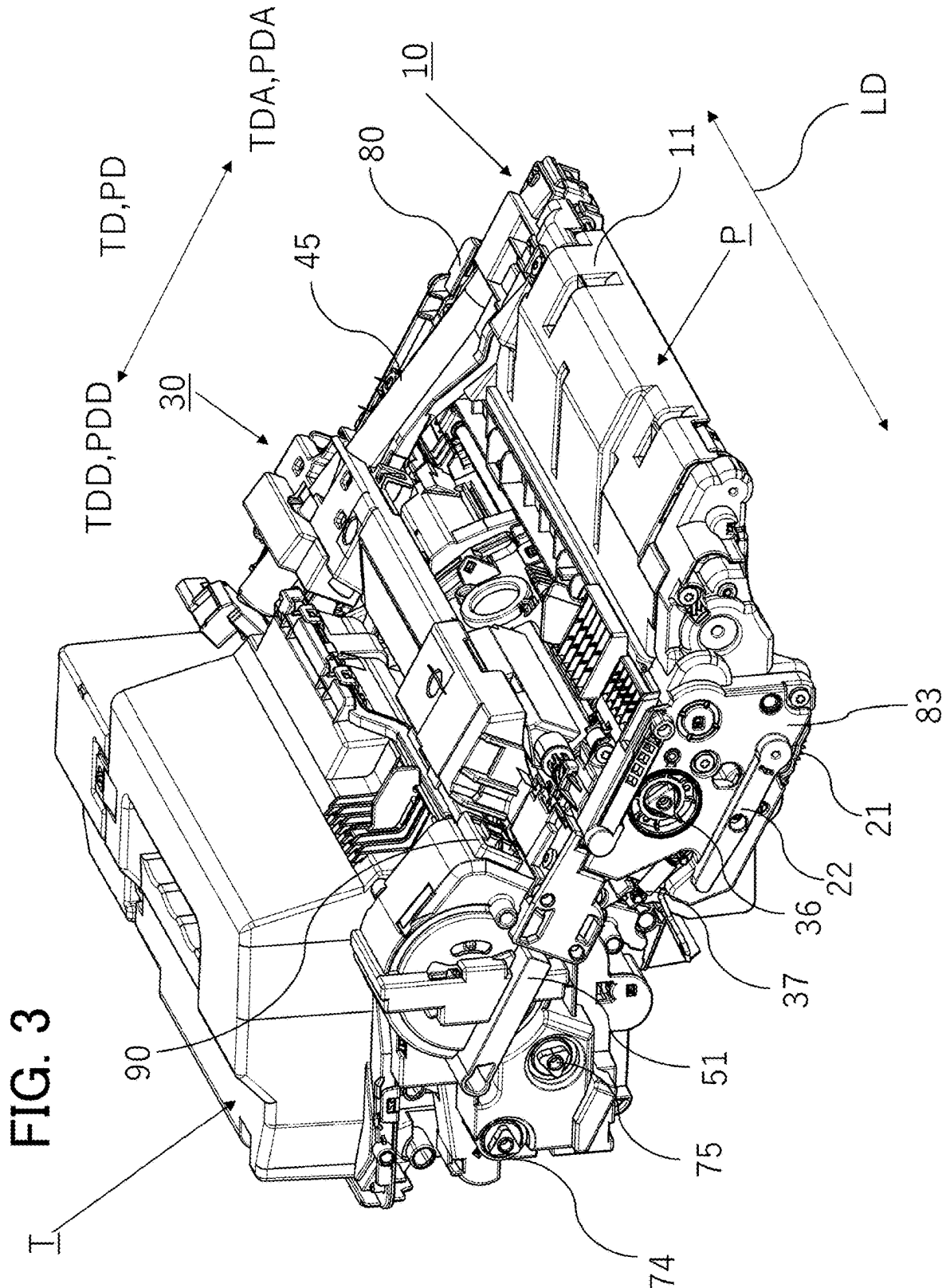


FIG. 2





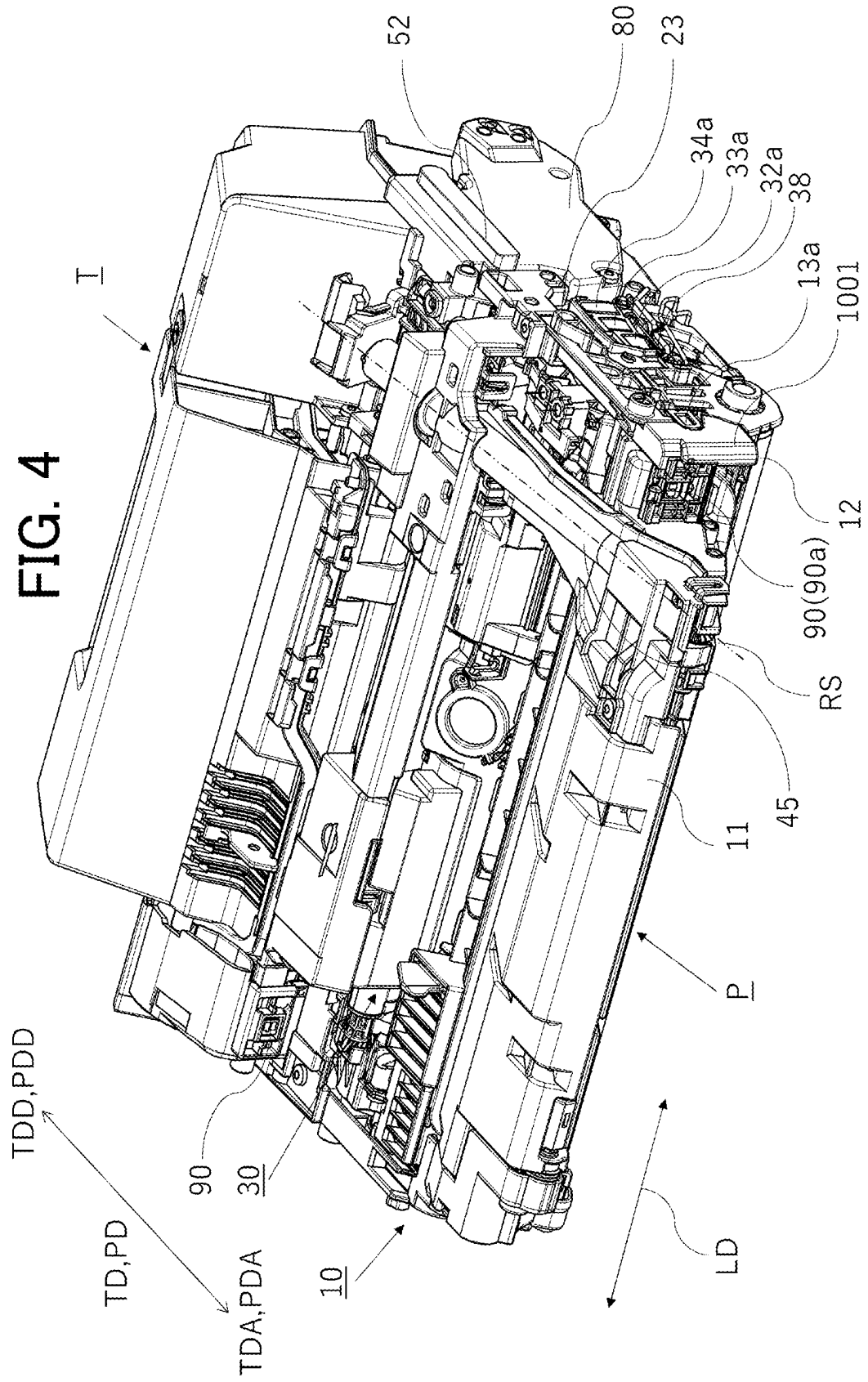
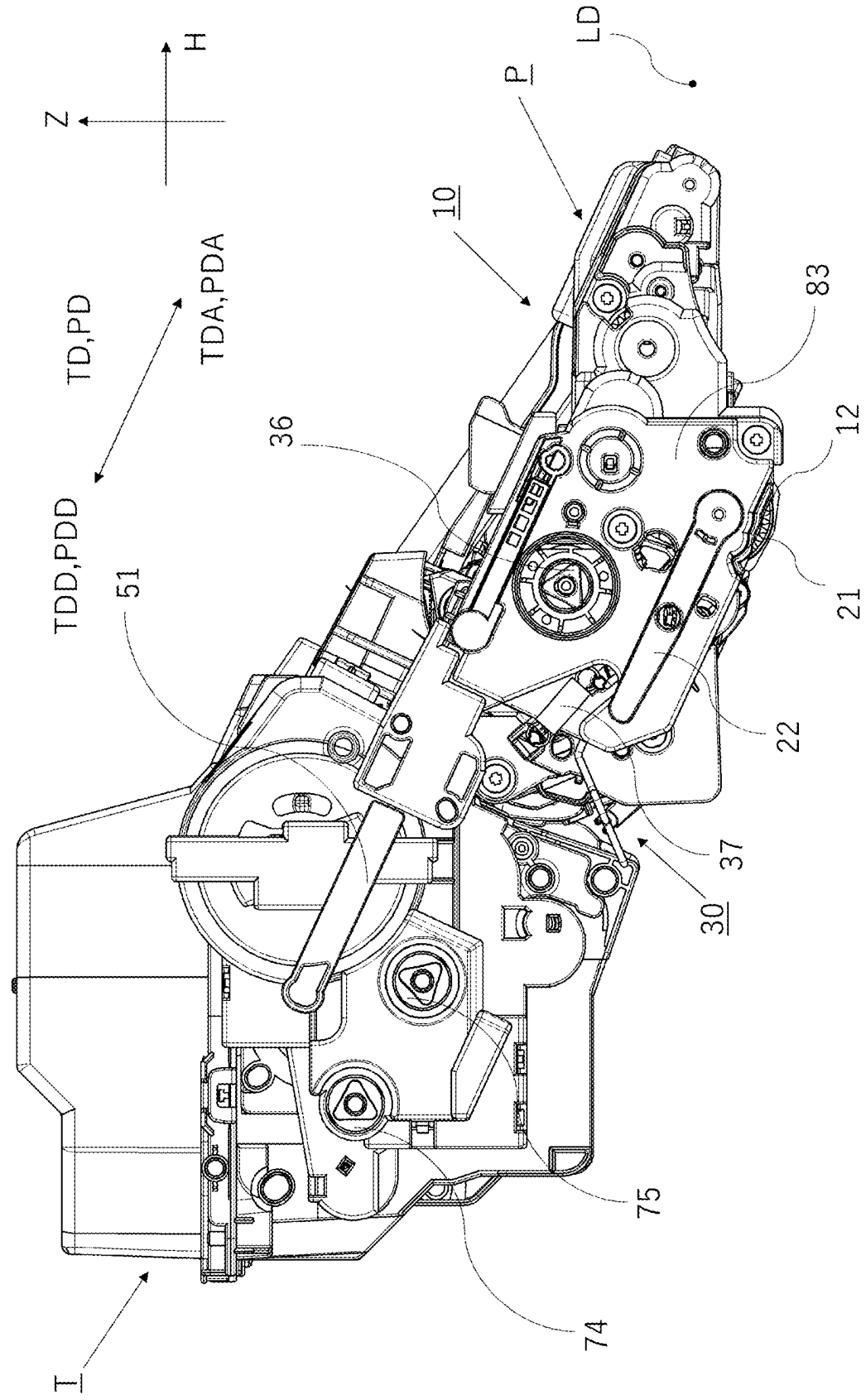


FIG. 5



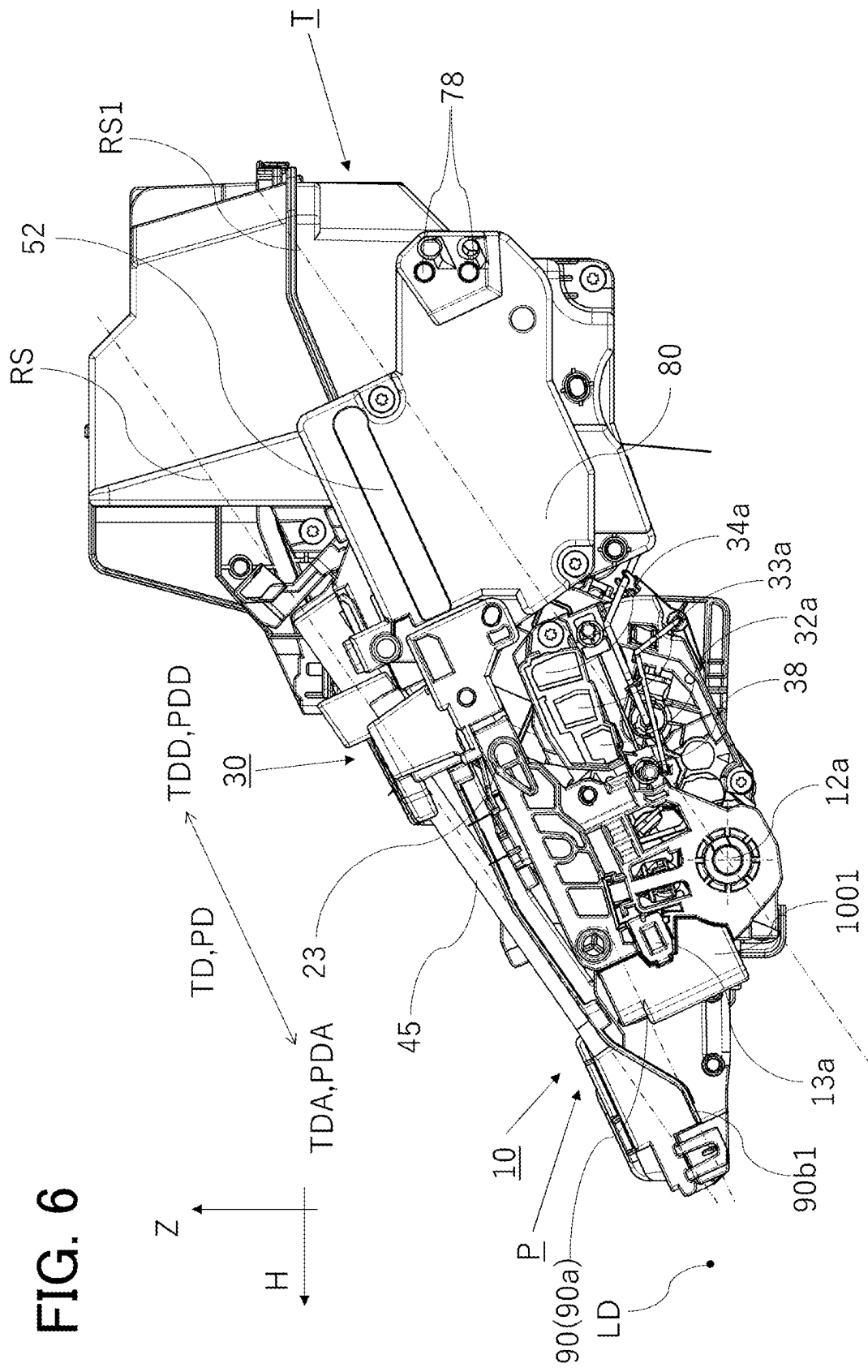
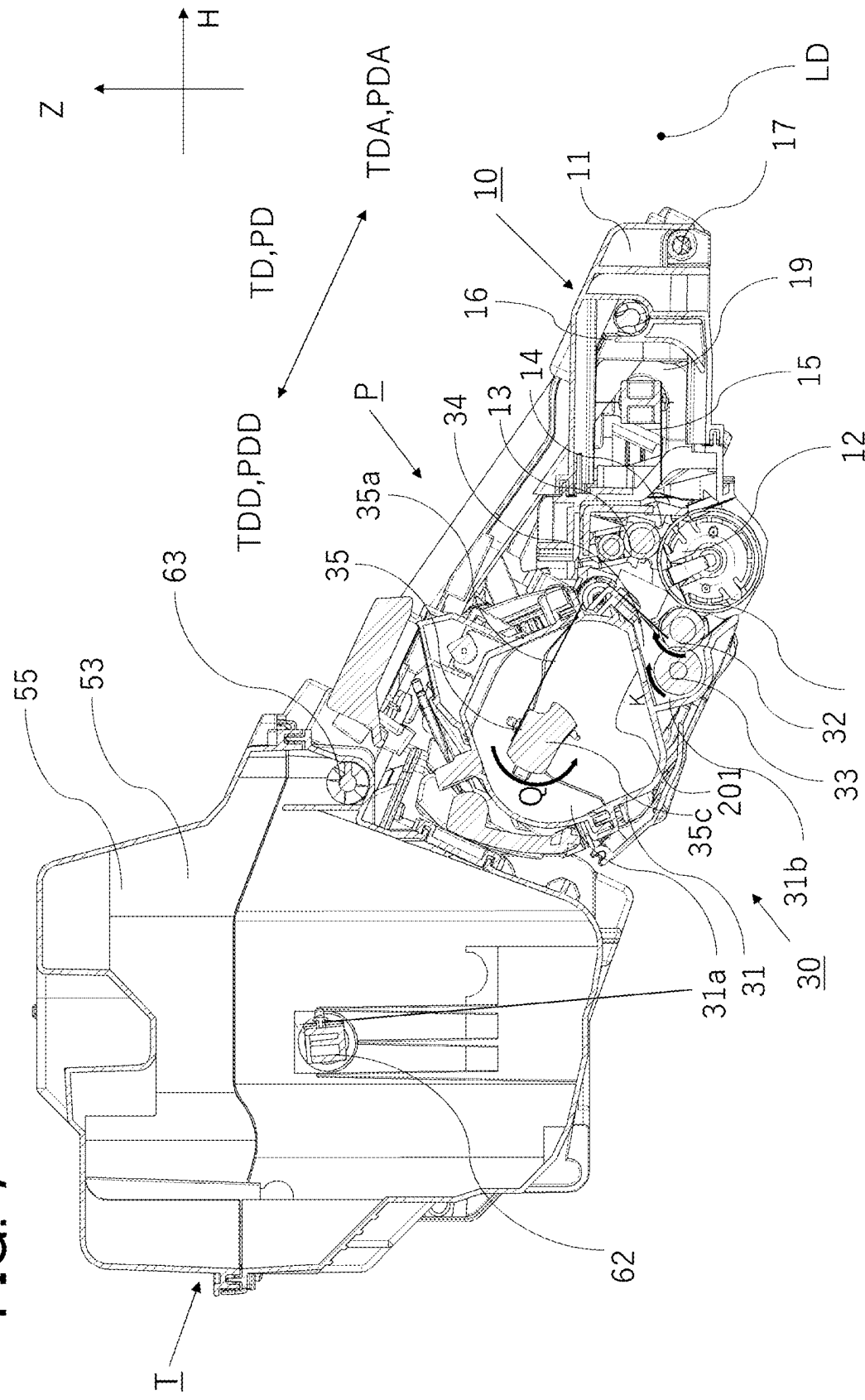
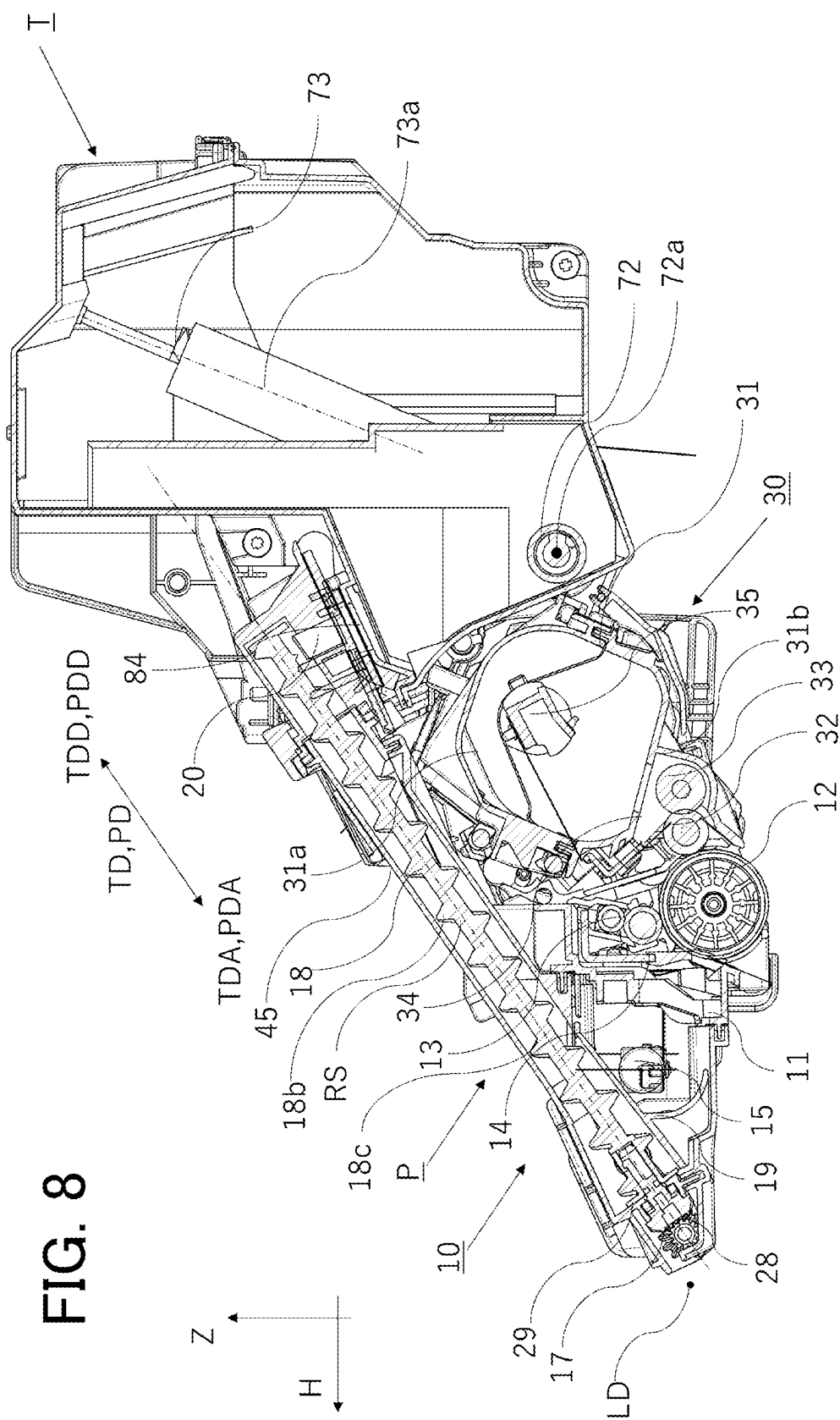


FIG. 6

FIG. 7





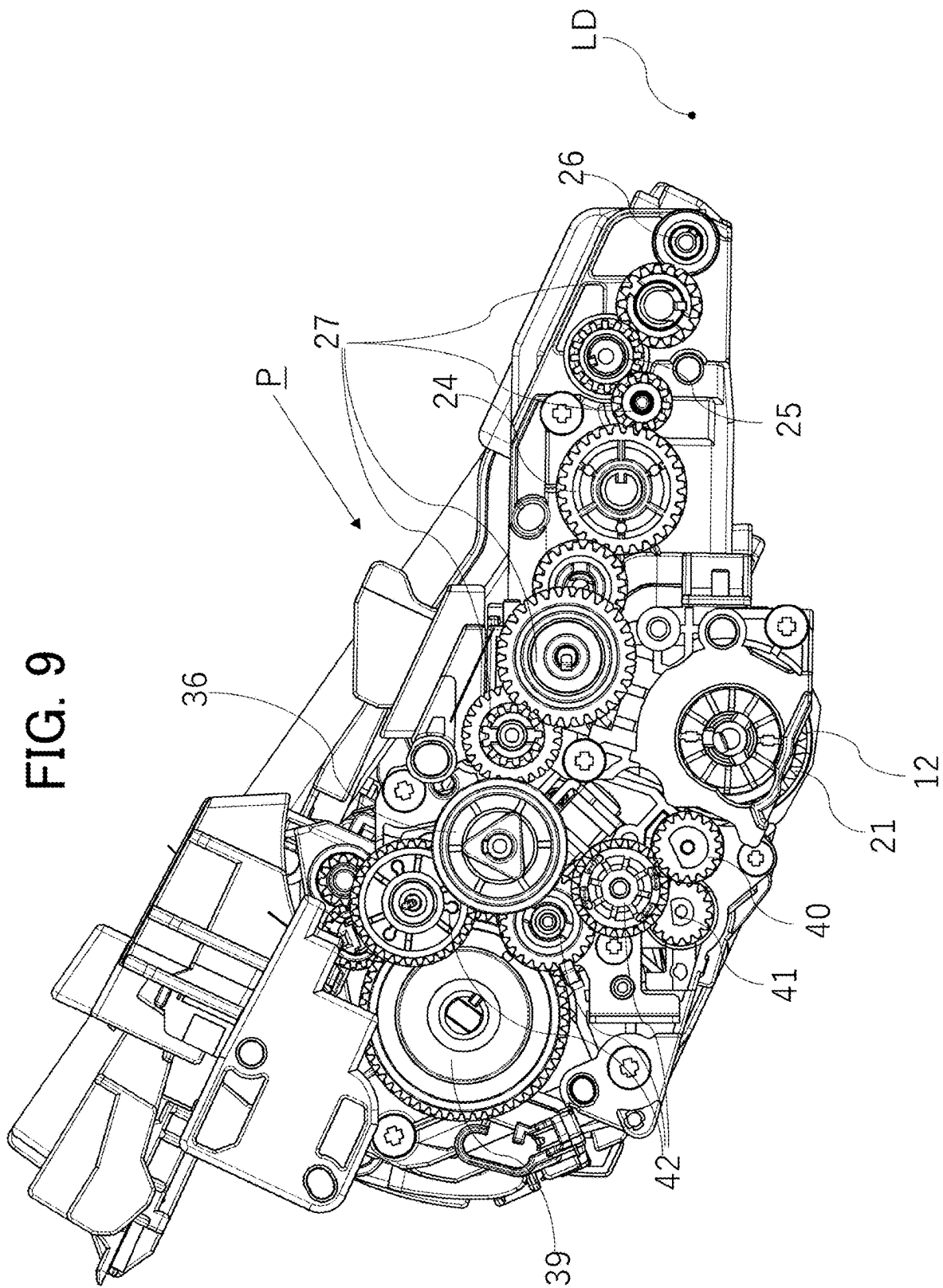
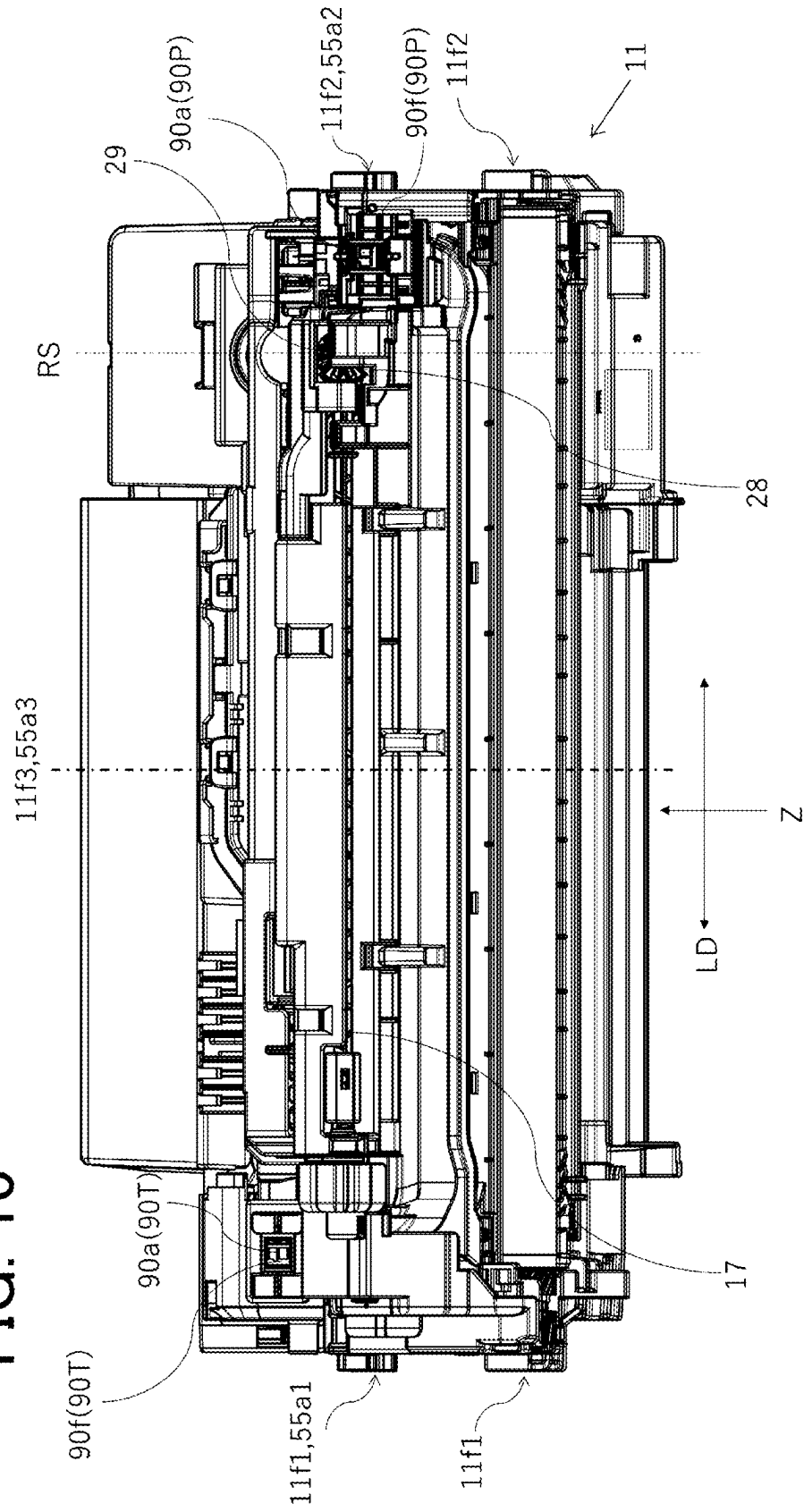
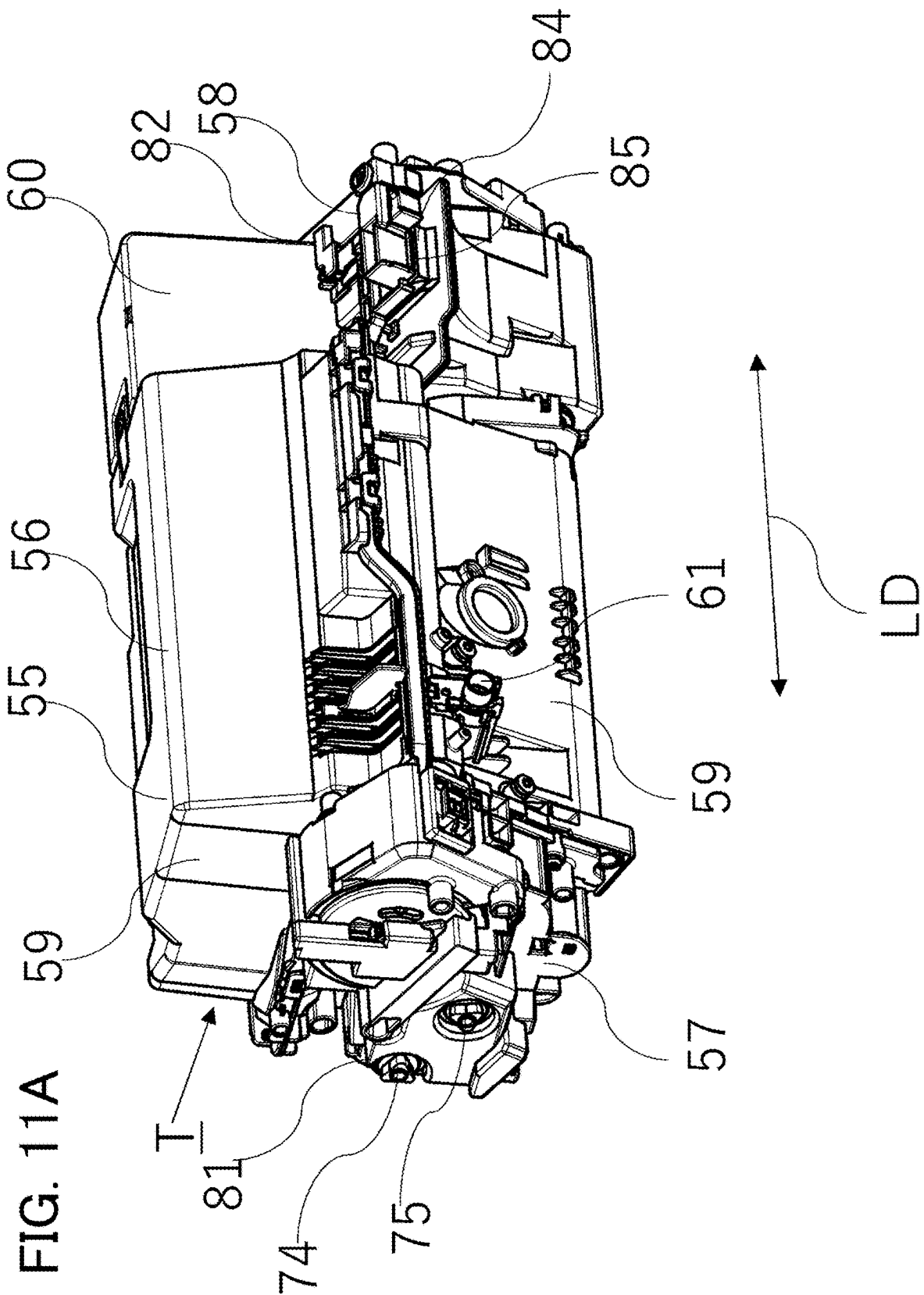
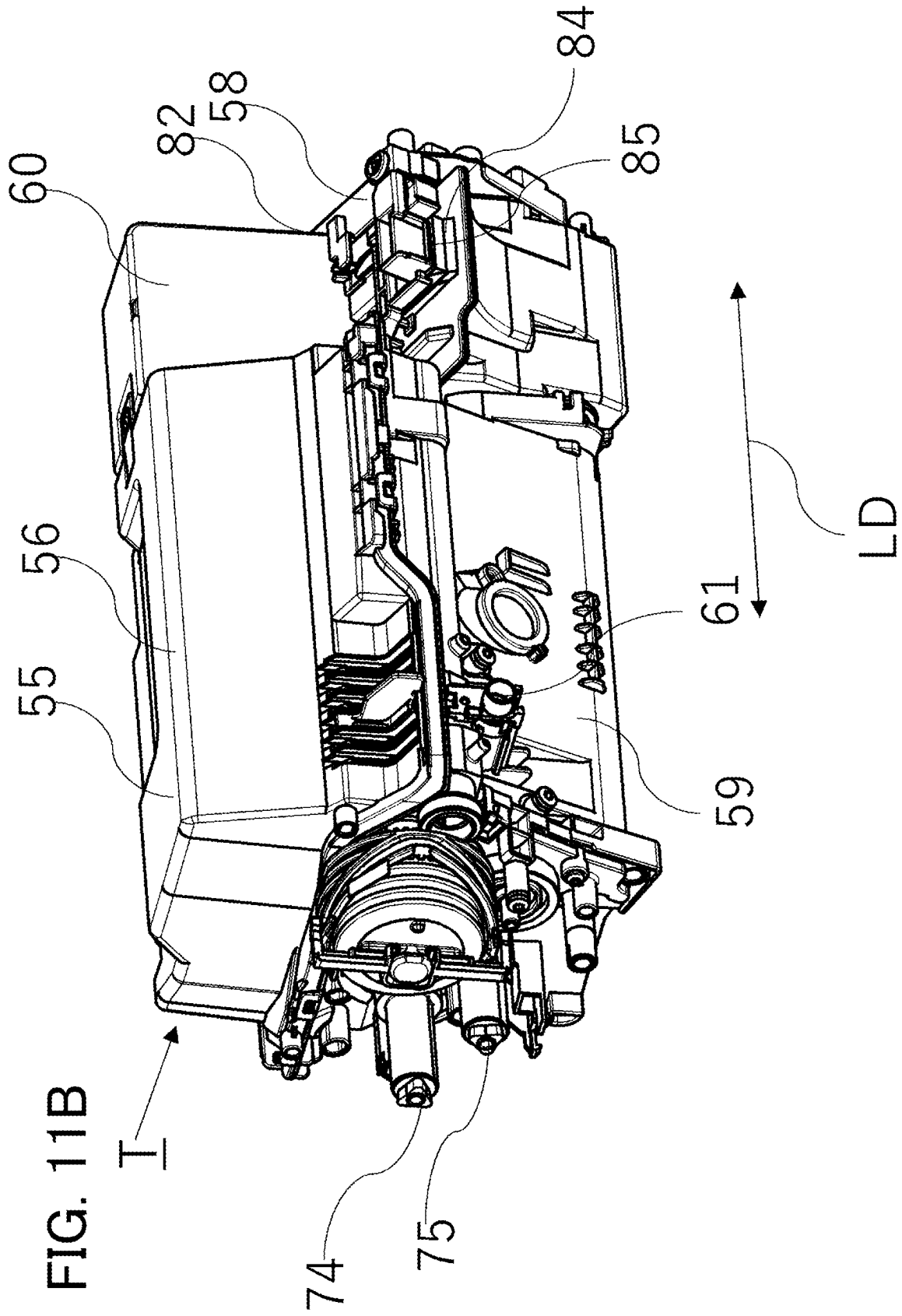
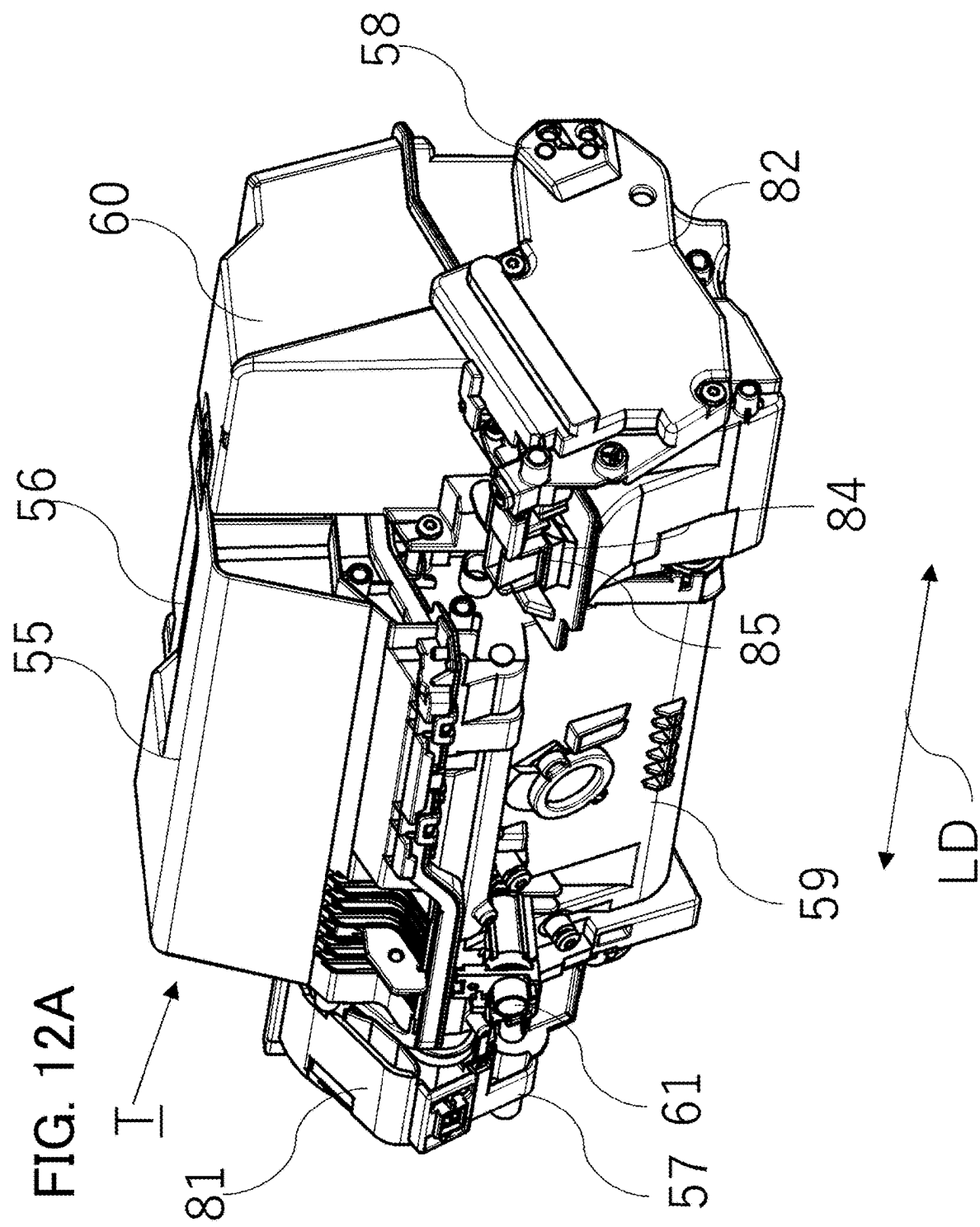


FIG. 10









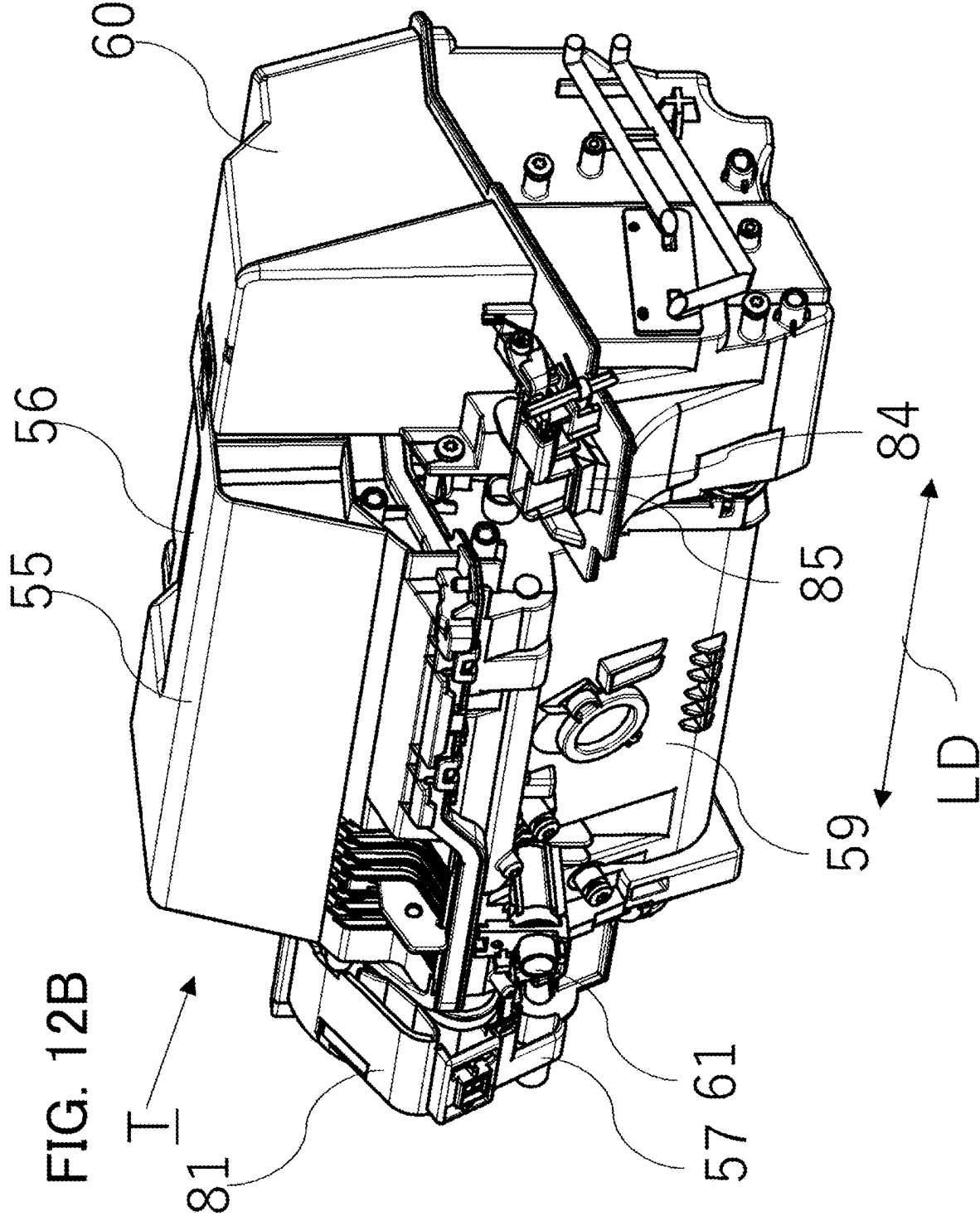


FIG. 13

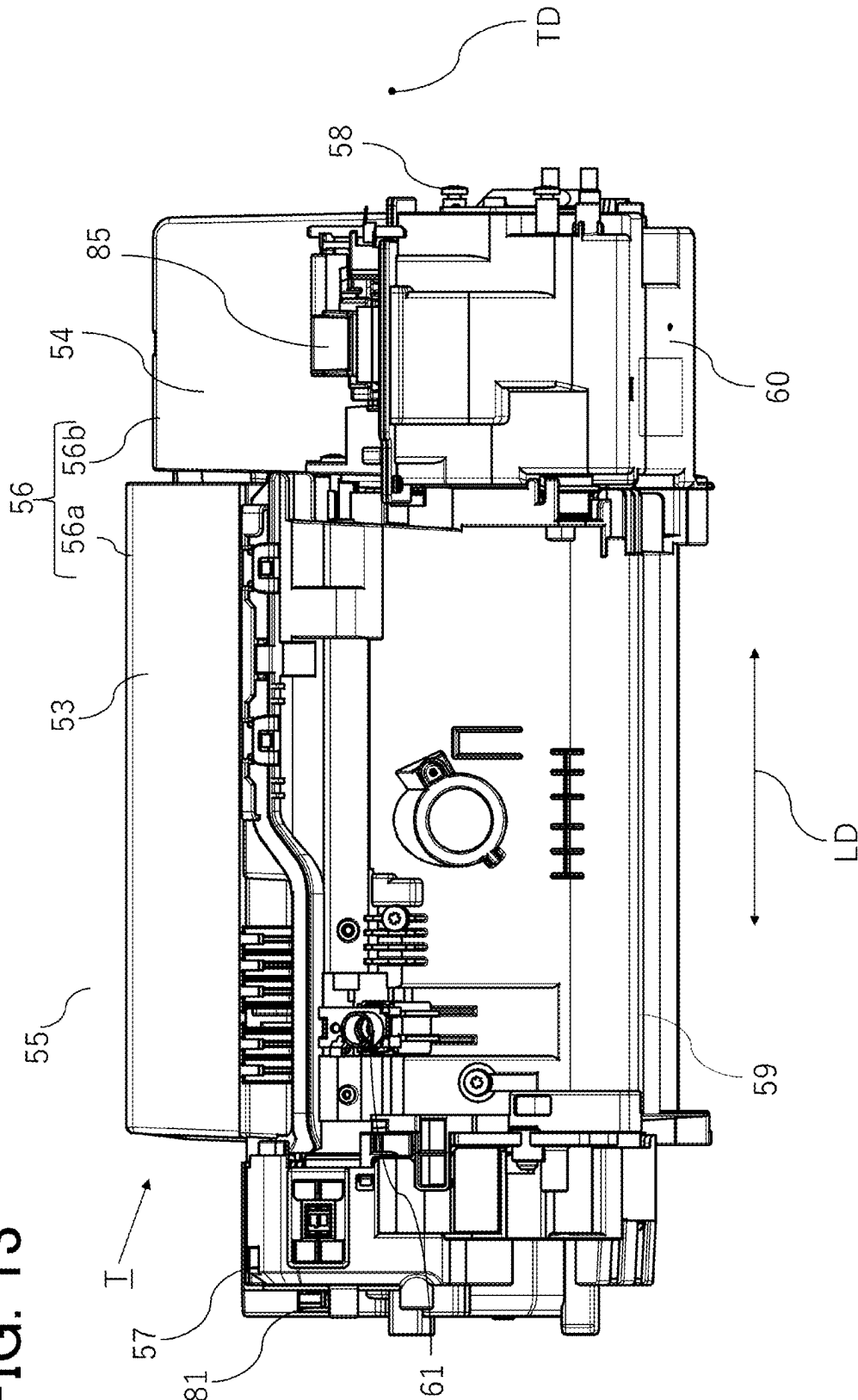


FIG. 14

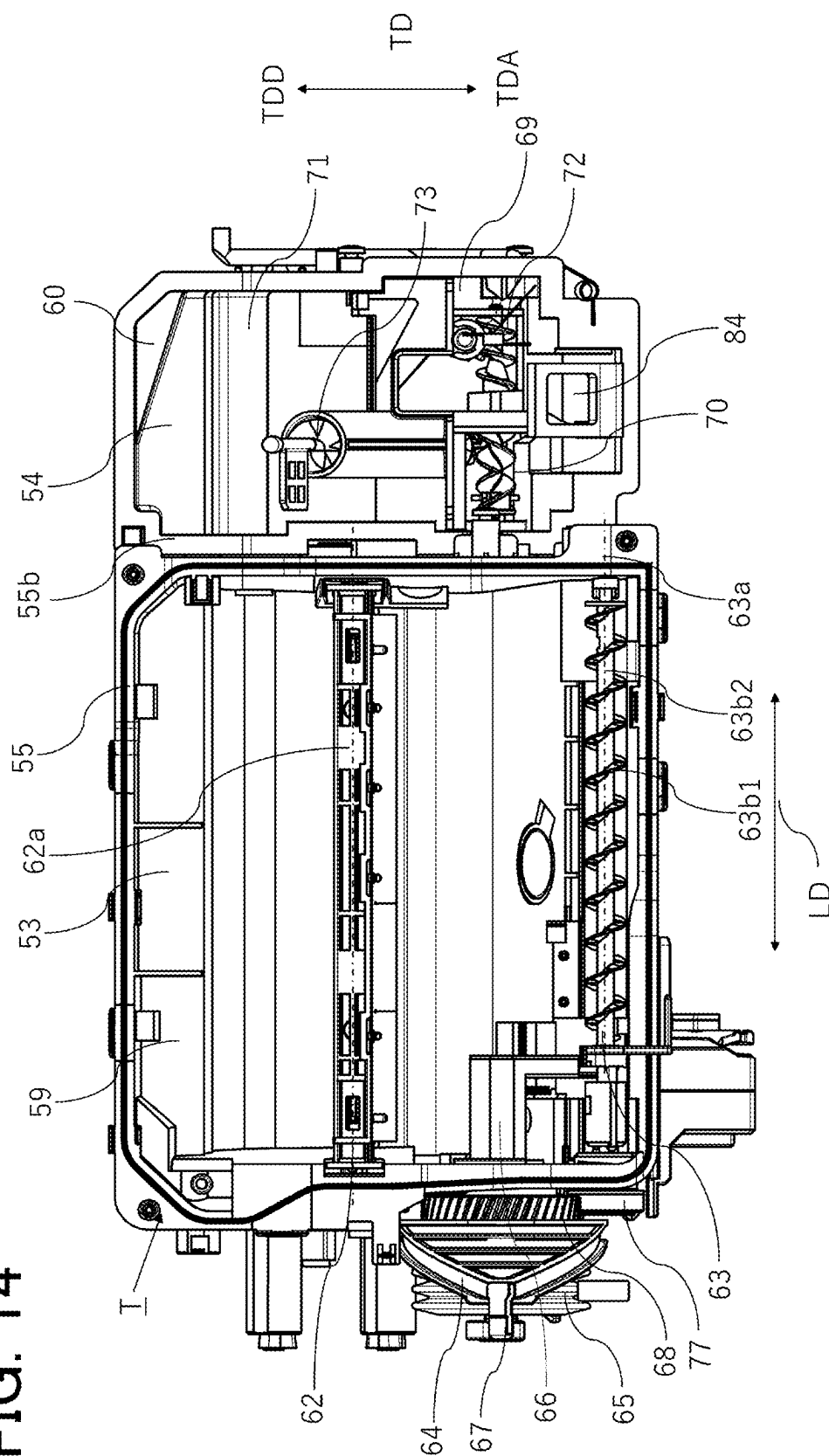
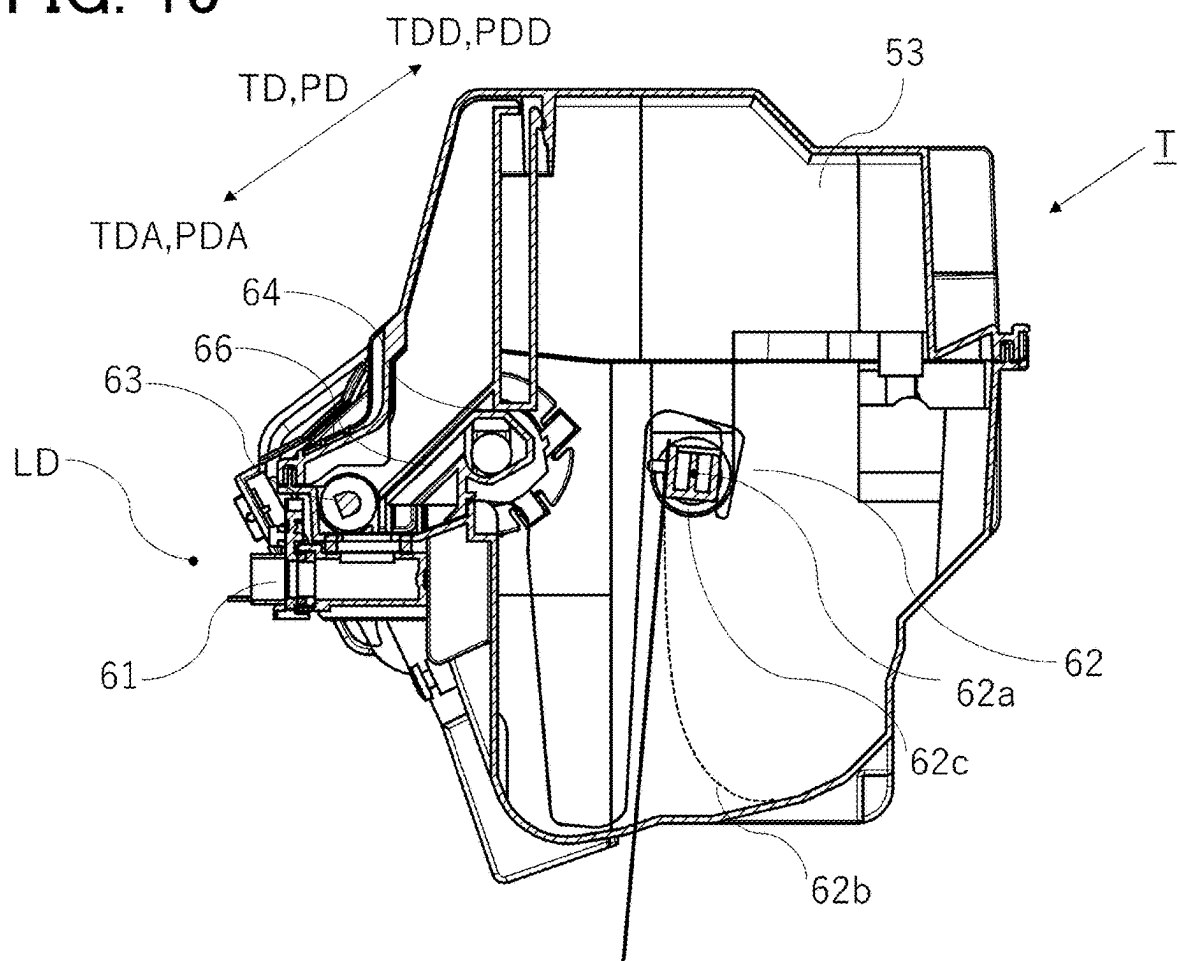
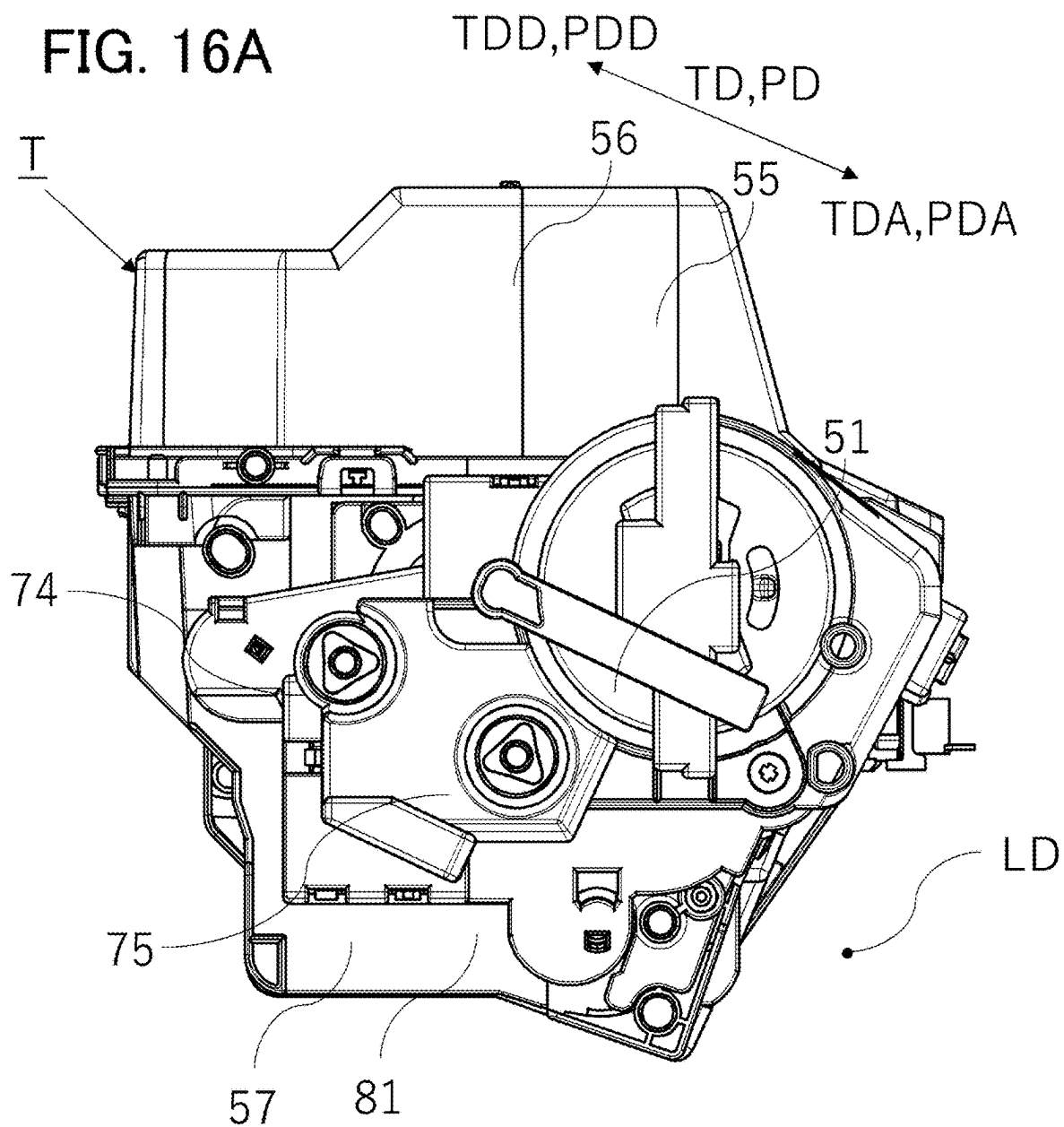


FIG. 15





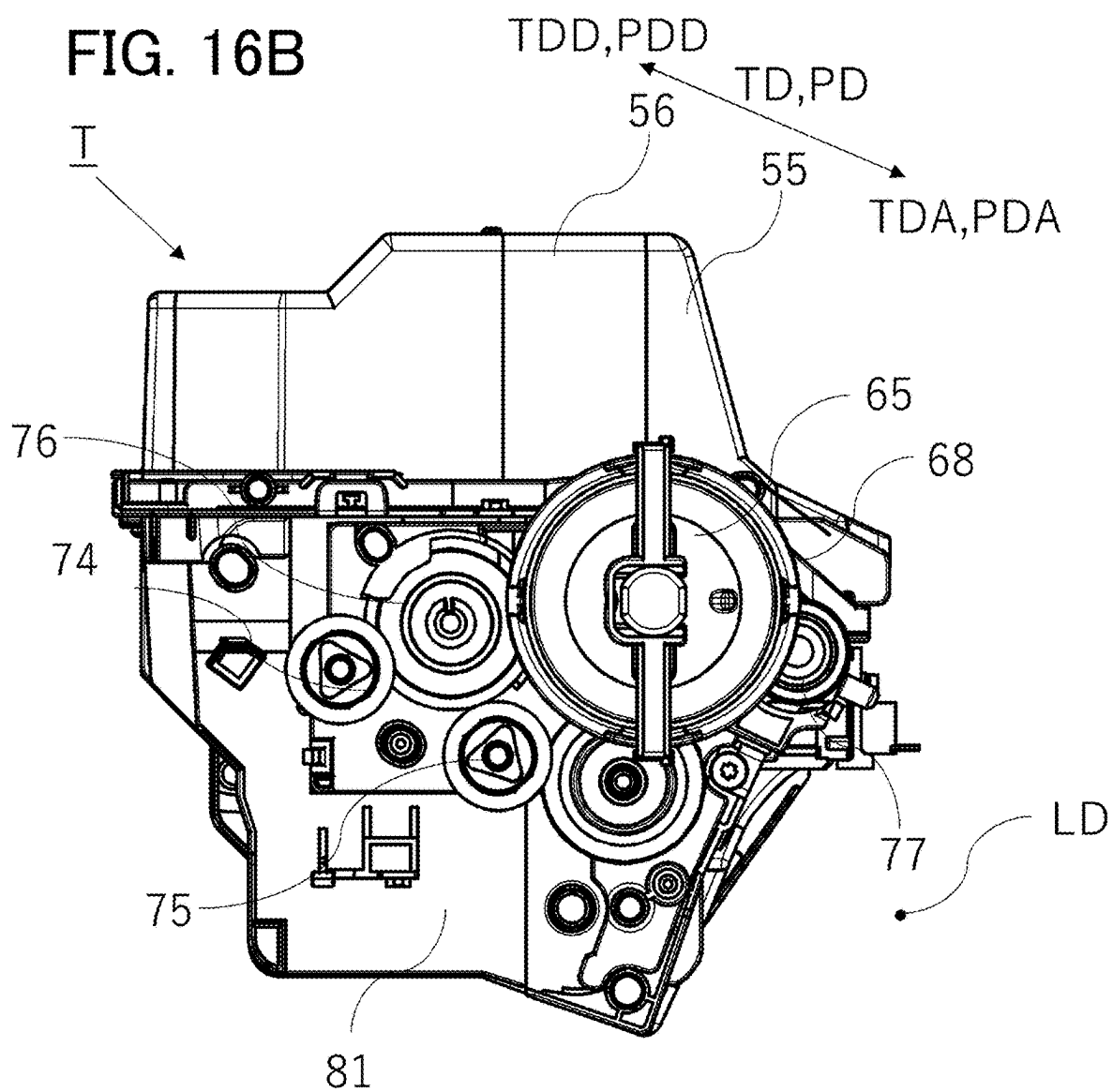


FIG. 17

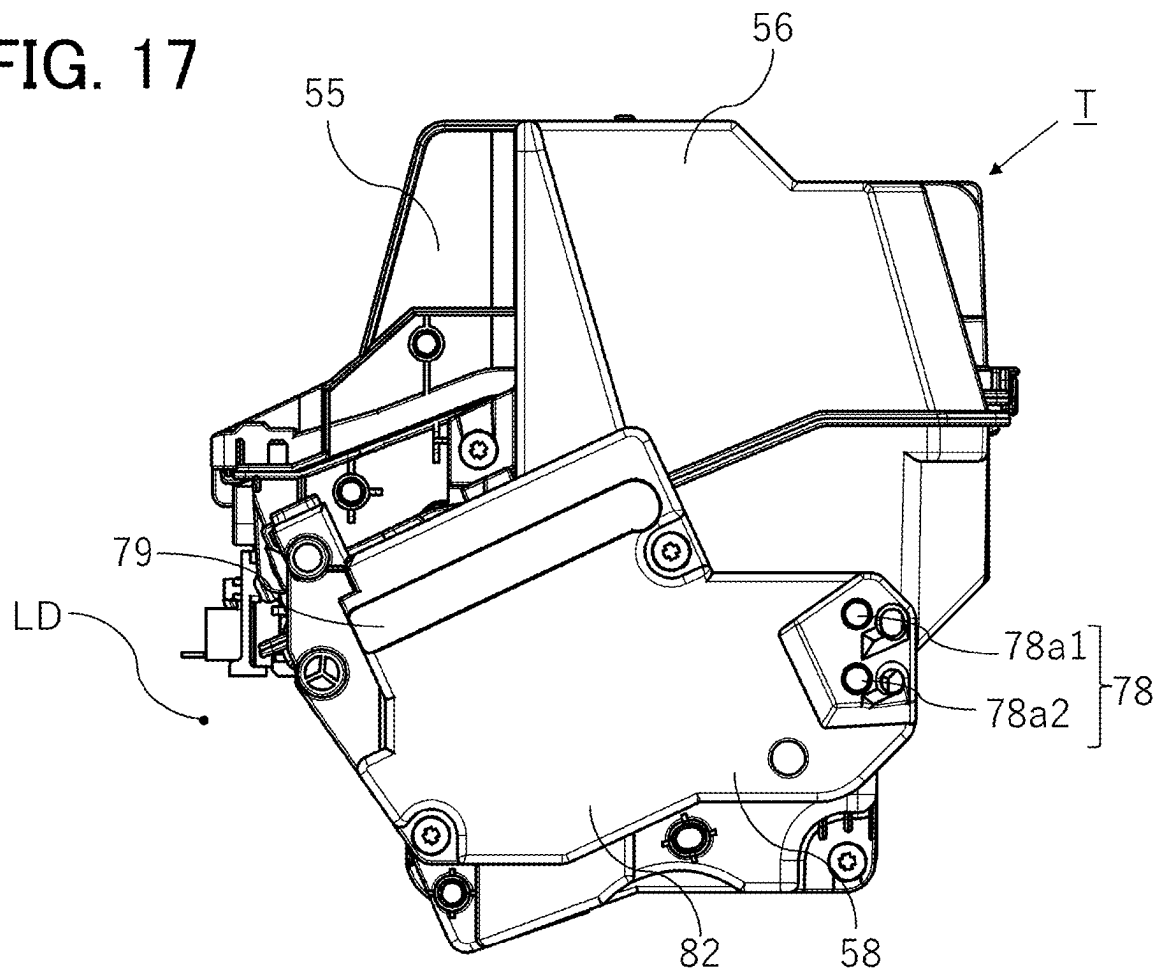


FIG. 18A

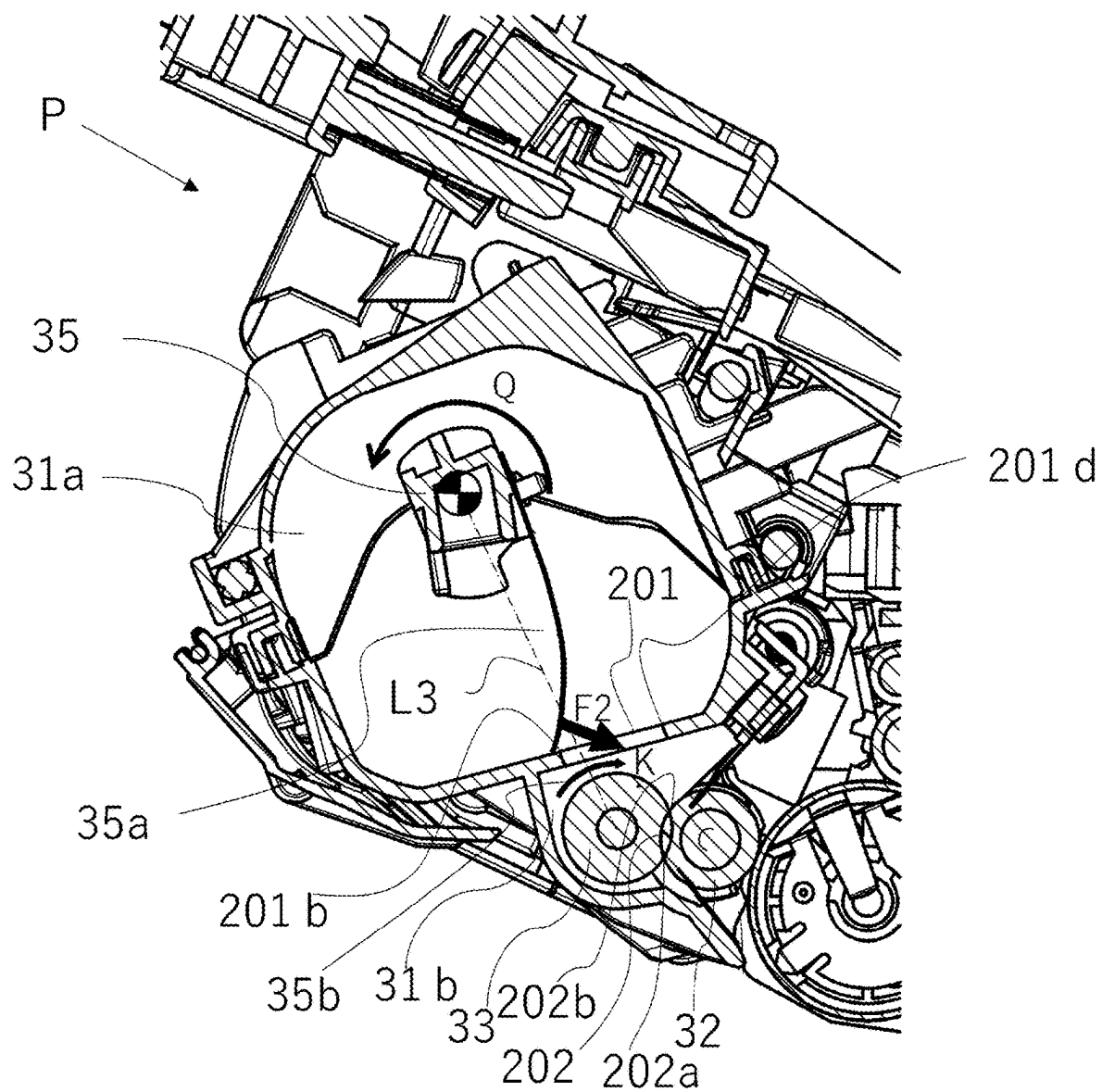


FIG. 18B

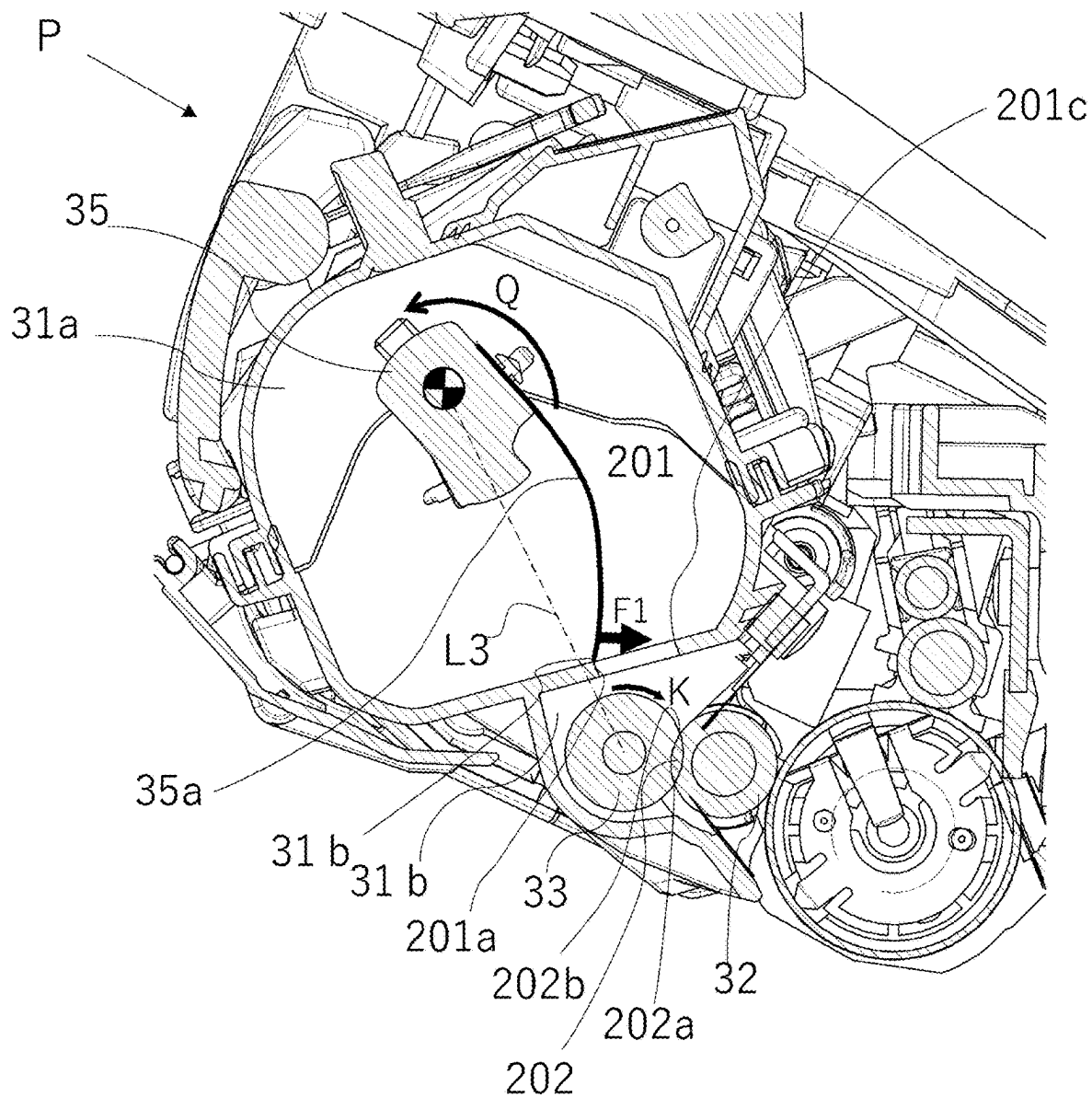


FIG. 19B

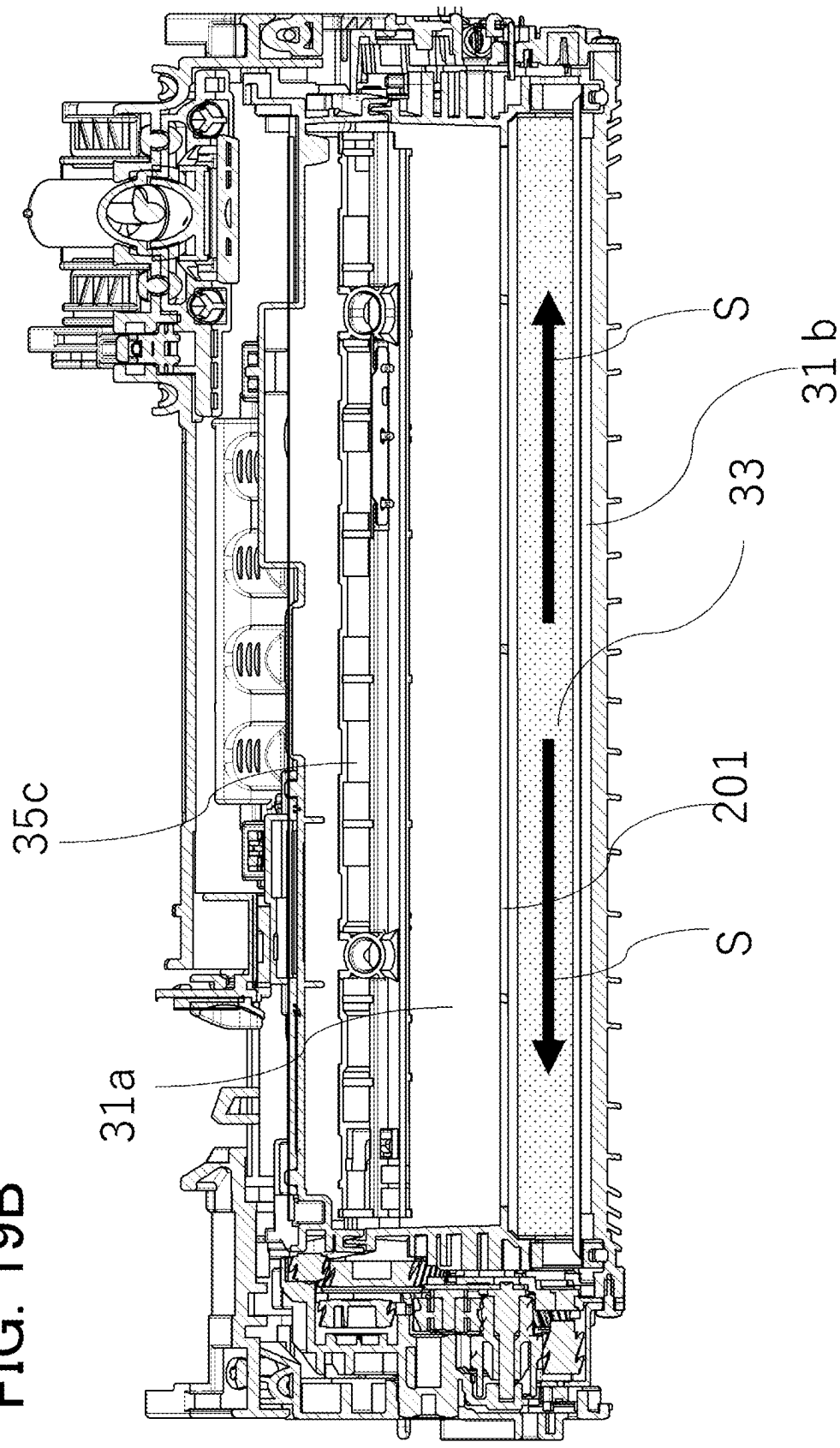
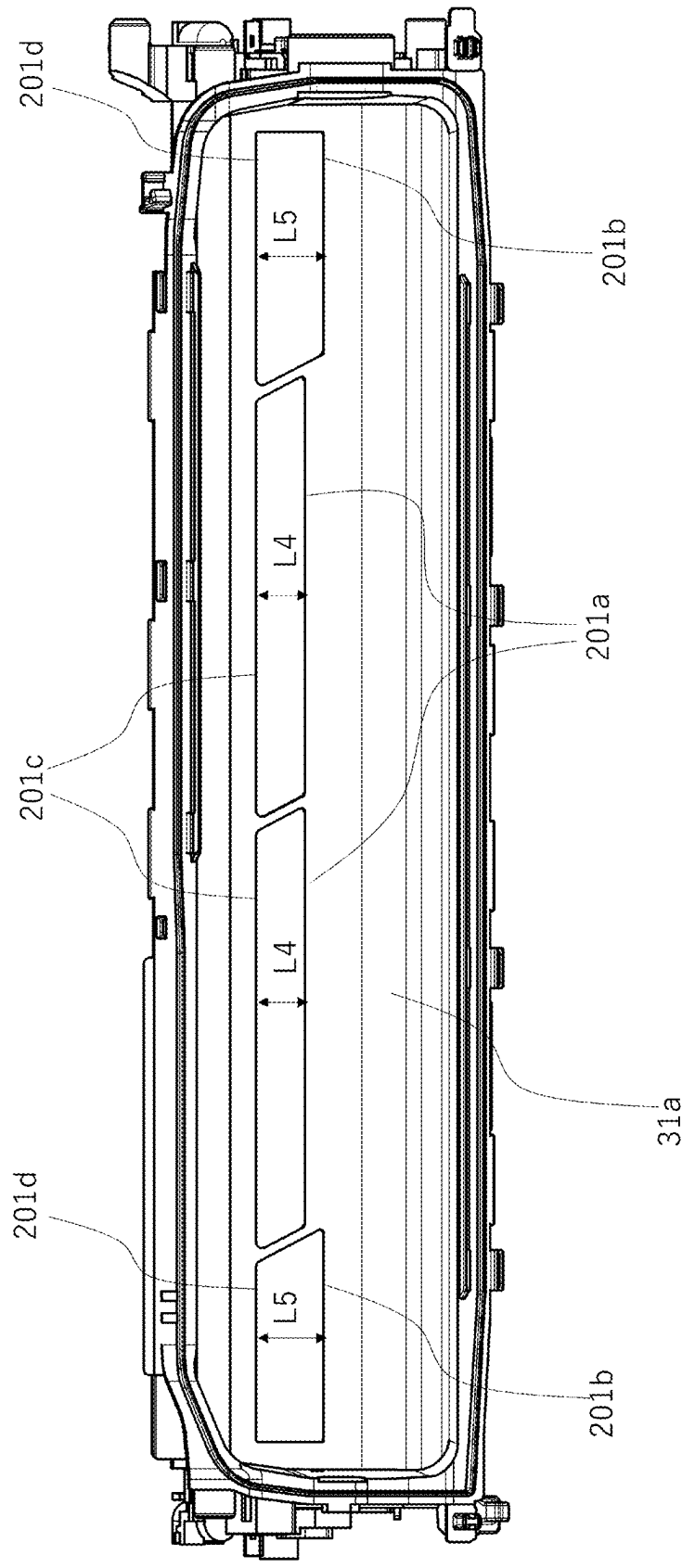
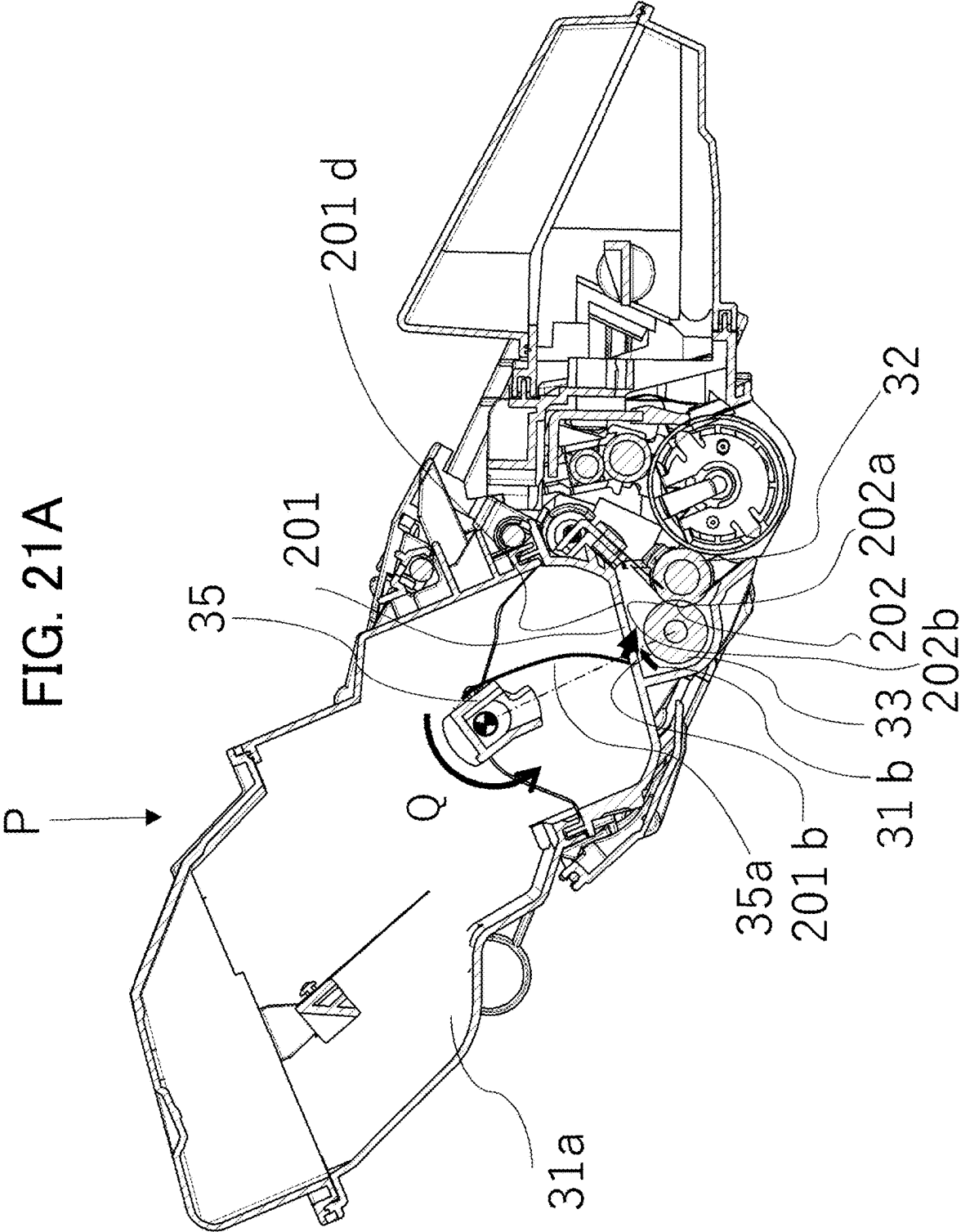


FIG. 20





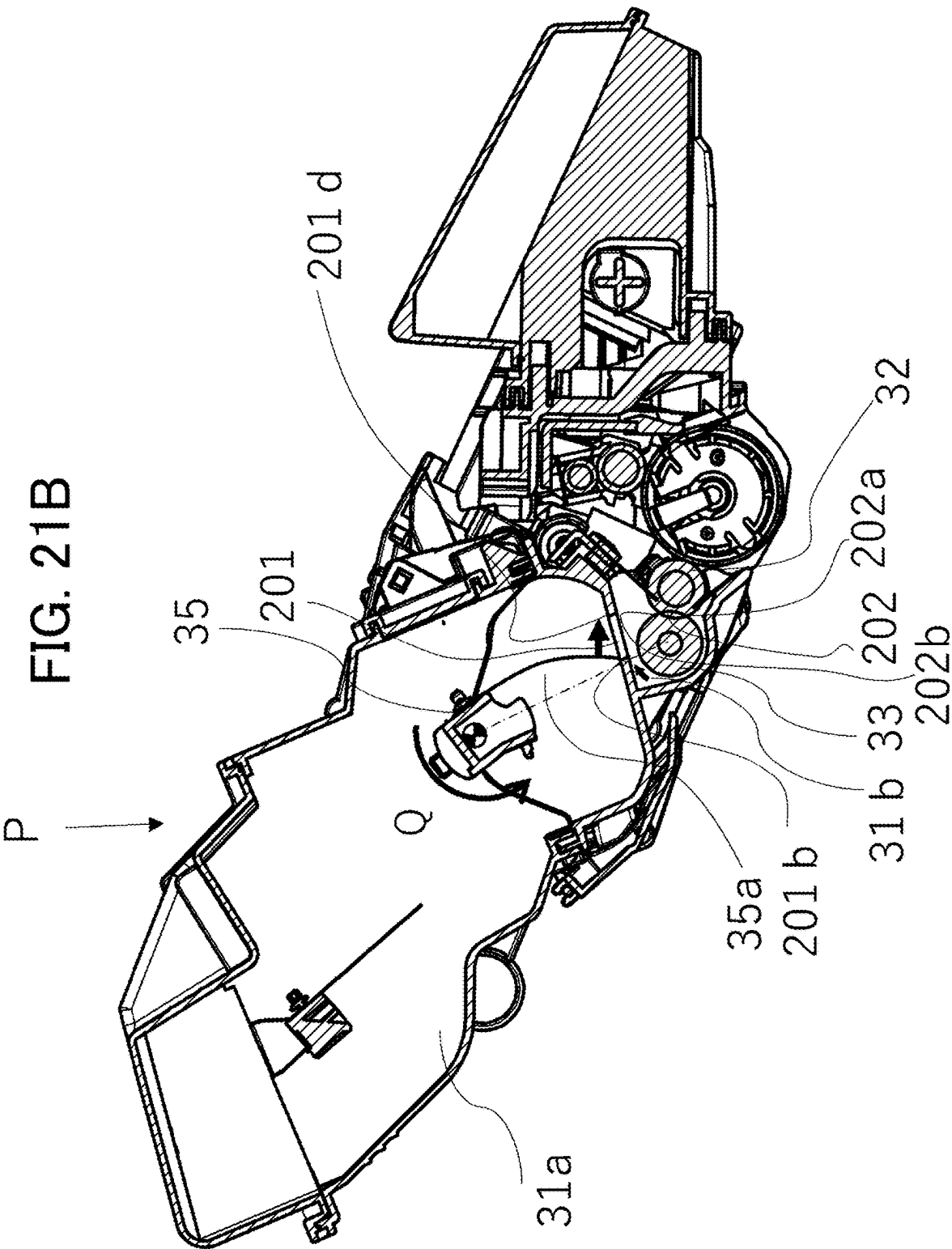


FIG. 22A

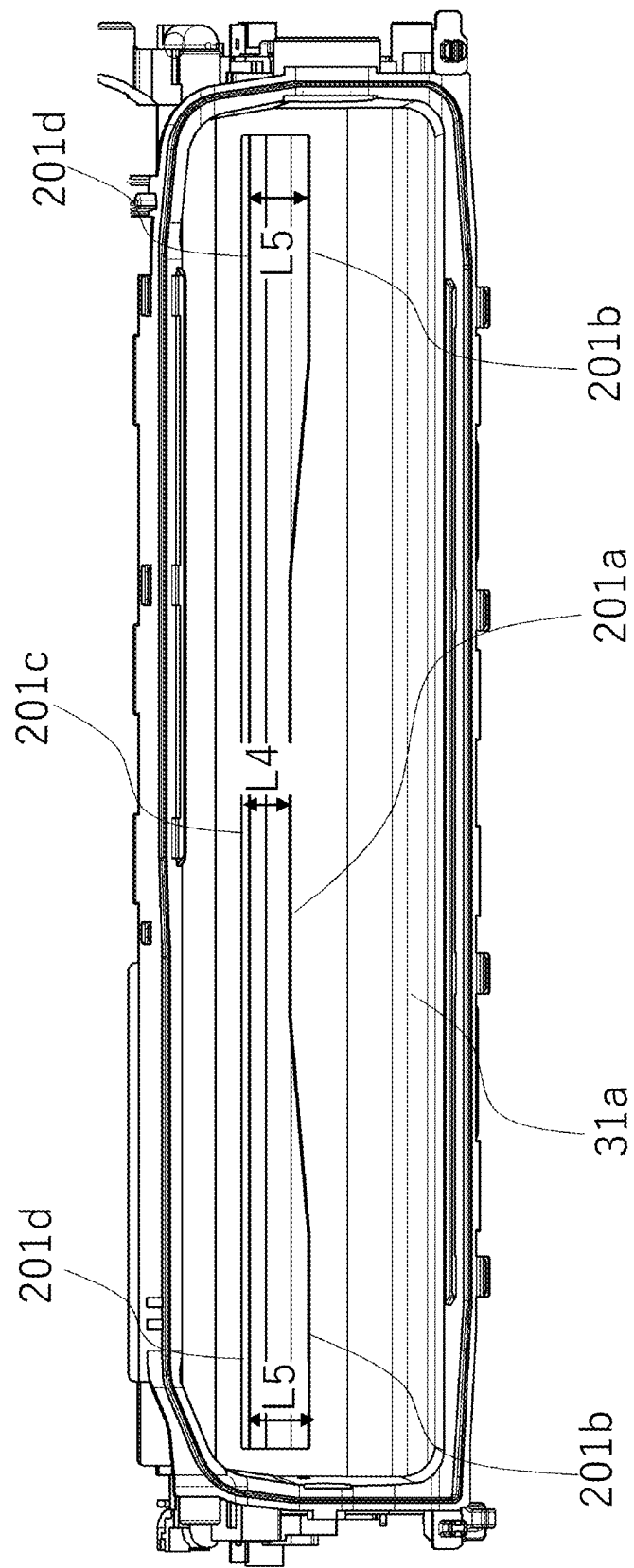


FIG. 22B

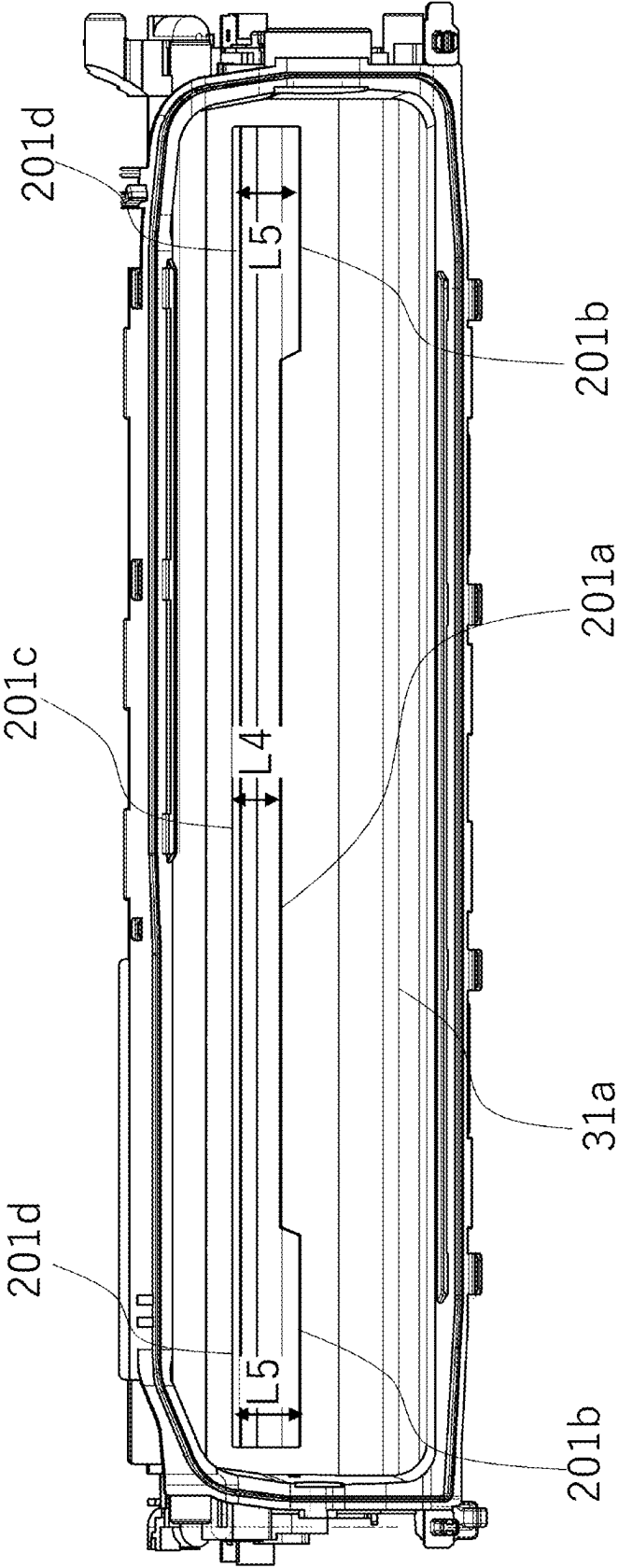
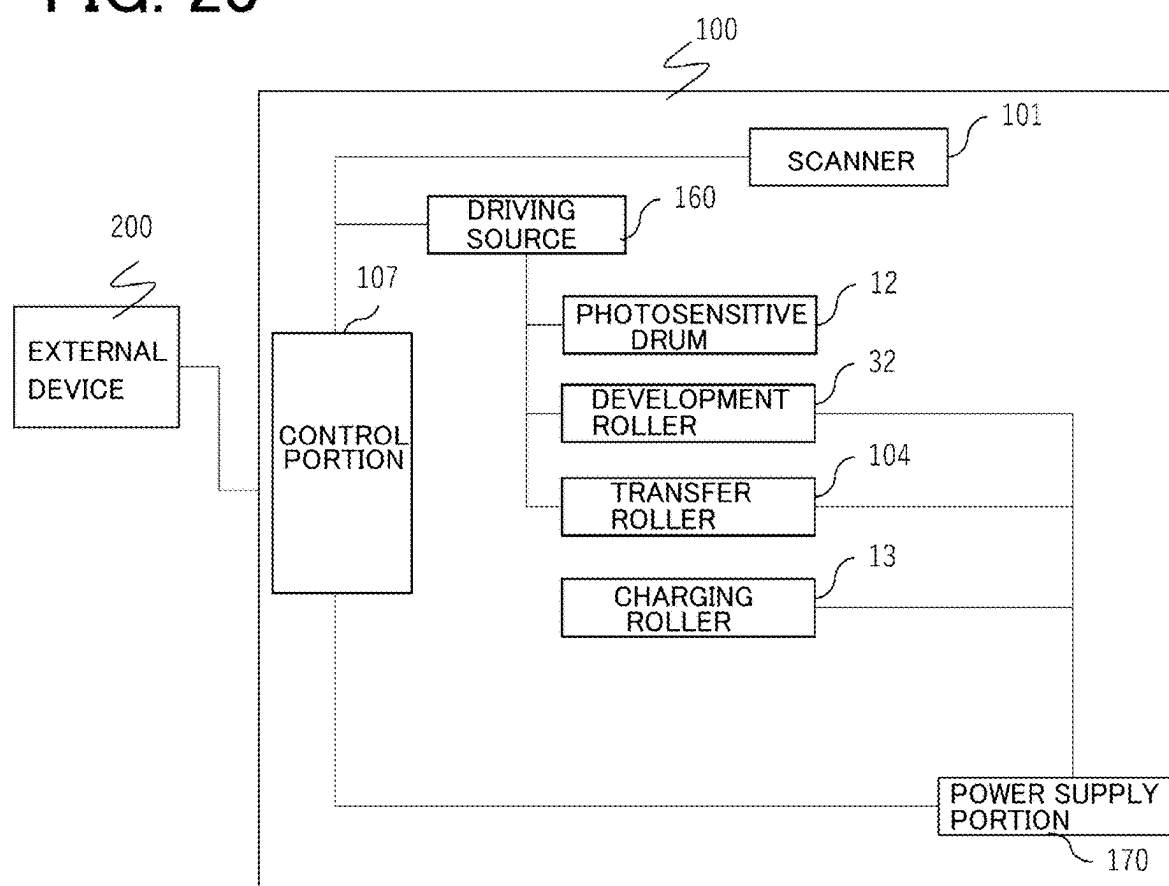


FIG. 23



DEVELOPING APPARATUS AND IMAGE FORMING APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a developing apparatus and an image forming apparatus.

Description of the Related Art

[0002] An image forming apparatus for forming an image on a recording medium using an electrophotographic image forming system is known. Examples of the image forming apparatus may include an electrophotographic copier, an electrophotographic printer (such as a LED printer or a laser beam printer), a facsimile device, and a word processor.

[0003] Japanese Patent Application Publication No. 2001-194883 discloses a development unit configuration having a development chamber in which a development roller and a supply roller are set, and a developer accommodation chamber for accommodating a developer (toner) therein. In Japanese Patent Application Publication No. 2001-194883, in the developer accommodation chamber provided at the top of the development chamber, a stirring member for stirring a developer is set.

[0004] When in the development unit as in Japanese Patent Application Publication No. 2001-194883, or a process cartridge including the same, the developer amount is increased due to an increase in volume, or a developer having a heavy specific gravity is used, the developer in the developer accommodation chamber becomes more likely to be pushed into the development chamber set at the lower part due to its own weight of the developer and the movement of the stirring member, resulting in a dense state of the developer in the development chamber. As a result, the poor circulation of the developer in the development chamber is caused, so that the functions of the development roller and the supply roller are impaired. This may result in the occurrence of a harmful effect on an image such as toner leakage. Under such circumstances, conventionally, the developer accommodation chamber has been set in a lying posture in the direction horizontal with the development chamber, thereby preventing its own weight of the developer from being imposed. However, when the developer accommodation chamber is set in a lying posture, an increase in size of the apparatus may be caused.

SUMMARY OF THE INVENTION

[0005] The present invention was completed in view of the foregoing problem. It is an object of the present invention to circulate a developer in a development unit properly.

[0006] The present invention provides a developing apparatus configured to comprise:

[0007] a developer bearing member for bearing a developer;

[0008] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;

[0009] a development chamber including the developer bearing member and the supply member arranged therein;

[0010] a developer accommodation chamber for accommodating the developer; and

[0011] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft,

wherein,

[0012] the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus,

[0013] the developer bearing member, the supply member, and the stirring member are rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center,

[0014] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and

[0015] in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

[0016] The present invention also provides an image forming apparatus configured to comprise:

[0017] a developer bearing member for bearing a developer;

[0018] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;

[0019] a development chamber including the developer bearing member and the supply member arranged therein;

[0020] a developer accommodation chamber for accommodating the developer;

[0021] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft; and

[0022] an image bearing member for developing an electrostatic latent image formed on a surface when the developer is supplied from the developer bearing member,

wherein,

[0023] the developer accommodation chamber is set at a top in a gravity direction of the development chamber,

[0024] the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction as a center,

[0025] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accom-

modation chamber to the development chamber along the longitudinal direction, and

- [0026] in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.
- [0027] The present invention also provides a developing apparatus configured to comprise:
 - [0028] a developer bearing member for bearing a developer;
 - [0029] a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;
 - [0030] a development chamber including the developer bearing member and the supply member arranged therein;
 - [0031] a developer accommodation chamber for accommodating the developer; and
 - [0032] a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft,
 wherein,
 - [0033] the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus,
 - [0034] the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center,
 - [0035] a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and
 - [0036] in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a width between the end on the upstream side and the end on the downstream side at a central portion in the longitudinal direction is narrower than a width between the end on the upstream side and the end on the downstream side at the ends in the longitudinal direction.
- [0037] In accordance with a configuration of the present invention, it is possible to circulate the developer in a development unit properly.
- [0038] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

- [0039] FIG. 1A is a cross sectional view showing a shape of an opening of a development unit;
- [0040] FIG. 1B is a cross sectional view showing a shape of an opening of a development unit;
- [0041] FIG. 2 is a schematic cross sectional view showing a configuration of a printer;
- [0042] FIG. 3 is a perspective view of a process cartridge and a toner cartridge;
- [0043] FIG. 4 is a perspective view of a process cartridge and a toner cartridge;
- [0044] FIG. 5 is a side view of a process cartridge and a toner cartridge;
- [0045] FIG. 6 is a side view of a process cartridge and a toner cartridge;
- [0046] FIG. 7 is a cross sectional view of a process cartridge and a toner cartridge;
- [0047] FIG. 8 is a cross sectional view of a process cartridge and a toner cartridge;
- [0048] FIG. 9 is a side view for illustrating a driving transmission path of a process cartridge;
- [0049] FIG. 10 is a view of a process cartridge and a toner cartridge as seen in the removing direction;
- [0050] FIG. 11A is a perspective view of a toner cartridge;
- [0051] FIG. 11B is a perspective view of a toner cartridge;
- [0052] FIG. 12A is a perspective view of a toner cartridge;
- [0053] FIG. 12B is a perspective view of a toner cartridge;
- [0054] FIG. 13 is a view of a toner cartridge as seen in the removing direction;
- [0055] FIG. 14 is a view showing an inside structure of a toner cartridge;
- [0056] FIG. 15 is a cross sectional view of a toner cartridge;
- [0057] FIG. 16A is a side view of a toner cartridge;
- [0058] FIG. 16B is a side view of a toner cartridge;
- [0059] FIG. 17 is a side view of a toner cartridge;
- [0060] FIG. 18A is a cross sectional view showing a position and a shape of an opening of a development unit;
- [0061] FIG. 18B is a cross sectional view showing a position and a shape of an opening of a development unit;
- [0062] FIG. 19A is a cross sectional view showing the movement of a developer in a process cartridge;
- [0063] FIG. 19B is a cross sectional view showing a movement of a developer in a process cartridge;
- [0064] FIG. 20 is a cross sectional view showing a shape of an opening of a development unit;
- [0065] FIG. 21A is a cross sectional view showing a shape of an opening of a development unit in accordance with another Example;
- [0066] FIG. 21B is a cross sectional view showing a shape of an opening of a development unit in accordance with a still other Example;
- [0067] FIG. 22A is a cross sectional view showing a shape of an opening of a development unit;
- [0068] FIG. 22B is a cross sectional view showing a shape of an opening of a development unit; and
- [0069] FIG. 23 is a block view showing a control configuration of an image forming apparatus.

DESCRIPTION OF THE EMBODIMENTS

- [0070] Below, referring to the accompanying drawings, preferred embodiments of the present invention will be described exemplarily in details. However, the dimensions,

the materials, and the shapes of the constituent components described in the following embodiments, the relative arrangement thereof, and the like should be appropriately changed according to the configuration of the apparatus to which the present invention is applied, and various conditions. Therefore, it is not intended that the scope of the present invention is limited only thereto unless otherwise specified.

EXAMPLE 1

Whole Outline of Printer

[0071] The basic configuration and the operation of a printer 100 as an image forming apparatus in accordance with the present Example will be described by reference to FIG. 2. FIG. 2 is a schematic cross sectional view showing the configuration of the printer 100 in accordance with the present Example. An arrow Z indicates the vertical direction, and an arrow H indicates a horizontal direction. As shown in FIG. 2, the printer 100 has an apparatus main body 100A, and a process cartridge P and a toner cartridge T removably mounted in the apparatus main body 100A as cartridges.

[0072] The apparatus main body 100A has a scanner 101 as an exposure device, a loading tray 102 for loading sheets S thereon, a paper feed roller 103, a transfer roller 104, a fixing portion 105, a discharge tray 106, and a control portion 107. Further, the process cartridge P and the toner cartridge T (developer cartridges) are removably mounted in the apparatus main body 100A.

[0073] The process cartridge P has a photosensitive drum 12, a charging roller 13 (charging member), a cleaning blade 14 (cleaning member), a drum unit 10, and a development unit 30 (developing apparatus) having a development roller 32, a supply roller 33, and a development frame body 31. A further configuration of the process cartridge P will be described later. The drum unit 10 has a drum frame body 11 described later. The photosensitive drum 12 is rotatably supported by the drum frame body 11. The development roller 32 is rotatably supported by the development frame body 31. Incidentally, a configuration in which the drum cartridge corresponding to the drum unit 10, and the development cartridge corresponding to the development unit 30 are respectively detachably set with respect to the image forming apparatus main body is also acceptable in place of such a configuration that the process cartridge P includes the drum unit 10 and the development unit 30.

[0074] The toner cartridge T can be mounted with respect to the process cartridge P. The toner cartridge T is configured so as to accommodate a toner as a developer, and to supply the toner to the development unit 30 of the process cartridge P. Although will be described by reference to the accompanying drawings later, the toner cartridge T has a toner transport member 62, a toner transport screw 63, and a toner frame body 55. The toner transport member 62 and the toner transport screw 63 are rotatably supported by the toner frame body 55.

[0075] The image formation operation onto a sheet S will be described. The control portion 107 of the printer 100 starts the image formation operation onto the sheet S on the basis of a signal received from an external device 200 shown in FIG. 23. The control portion 107 includes an information processing device having arithmetic resources such as a processor and a memory, a processing circuit, and the like. As shown in a block view of FIG. 23, the control portion 107

acquires information from each constituent element of the printer 100 according to the program stored in a memory, or an user instruction using the external device 200, and performs control by a control signal. The printer 100 includes a driving source 160 having a motor, or the like, and a power supply portion 170 for supplying an electric power such as a driving voltage to each constituent element.

[0076] First, the photosensitive drum 12 is rotated by the driving source 160 of the apparatus main body 100A. The charging roller 13 is rotated following the photosensitive drum 12 with the charging roller 13 applied with a charging voltage by the power supply portion 170. As a result, the surface of the photosensitive drum 12 is uniformly charged. On the basis of image information, the scanner 101 applies a laser toward the charged surface of the photosensitive drum 12, so that an electrostatic latent image is formed on the surface of the photosensitive drum 12.

[0077] When the development roller 32 is applied with a development voltage by the power supply portion 170, a toner is supplied to the photosensitive drum 12 from the development roller 32. As a result, an electrostatic latent image is developed as a toner image. Subsequently, the rotation of the photosensitive drum 12 transports the toner image formed at the photosensitive drum 12 toward the transfer portion formed between the transfer roller 104 and the photosensitive drum 12.

[0078] On the other hand, a sheet S is fed from the loading tray 102 by the paper feed roller 103. The sheet S is transported to the transfer portion in accordance with the timing at which the toner image formed at the photosensitive drum 12 reaches the transfer portion.

[0079] A transfer voltage is applied to the transfer roller 104 from the power supply portion 170, so that the toner image is transferred from the photosensitive drum 12 to the sheet S. The residual toner not transferred to the sheet S is removed from the surface of the photosensitive drum 12 by a cleaning blade 14.

[0080] The sheet S onto which the toner image has been transferred is transported toward the fixing portion 105. When the sheet S passes through the fixing portion 105, the toner image is heated and pressurized by a fixing portion 105, thereby to be fixed onto the sheet S.

[0081] The printer 100 in accordance with the present Example can carry out double-sided printing in which an image formation operation is performed on the front surface and the back surface of the sheet S. When an image is formed on only the front surface of the sheet S, the sheet S that has passed through the fixing portion 105 is discharged to the discharge tray 106. On the other hand, when double-side printing is carried out, the sheet S with the toner image fixed on the surface thereof is transported in an arrow M direction through a double-side transport path 81, and reaches the transfer portion again. As a result, a toner image is formed on the back surface of the sheet S. Subsequently, the sheet S passes through the fixing portion 105, and is discharged to the discharge tray 106.

[0082] Attachment and Removal of Process Cartridge and Toner Cartridge

[0083] Referring to FIGS. 2, 3, 4, 5, 6, and 10, a description will be given to attachment and removal of a process cartridge P and a toner cartridge T in accordance with the present Example with respect to the apparatus main body 100A.

[0084] FIGS. 3 and 4 are each a perspective view of the process cartridge P and the toner cartridge T. FIGS. 5 and 6 are each a side view of the process cartridge P and the toner cartridge T. The photosensitive drum 12 is rotatable around the rotation axis (first axis) 12a shown in FIG. 6. The direction of extension of the rotation axis 12a is referred to as the rotation axis direction (axis direction) of the photosensitive drum 12.

[0085] FIG. 3 is a perspective view of the process cartridge P and the toner cartridge T as seen from the driving side. FIG. 4 is a perspective view of the process cartridge P and the toner cartridge T as seen from a non-driving side. FIG. 5 is a side view of the driving side of the process cartridge P and the toner cartridge T as seen in the rotation axis direction of the photosensitive drum 12. FIG. 6 is a side view of the non-driving side of the process cartridge P and the toner cartridge T as seen in the rotation axis direction of the photosensitive drum 12.

[0086] As shown in FIG. 2, the printer 100 has a door (opening/closing member) 100B that covers an opening 100C of the apparatus main body 100A. The door 100B is attached rotatably around the rotation axis 100D with respect to the apparatus main body 100A. The door 100B is configured movably in an arrow J direction from the closing position covering the opening 100C to the opening position exposing the opening 100C. With the door 100B present at the opening position, the attachment of the process cartridge P and the toner cartridge T to the apparatus main body 100A, and the removal of the process cartridge P and the toner cartridge T from the apparatus main body 100A through the opening 100C are allowed.

[0087] FIG. 10 is a view of the process cartridge P and the toner cartridge T as seen in the removal direction PDD. As shown in FIG. 10, with regard to the direction of the rotation axis 12a of the photosensitive drum 12, the drum frame body 11 has a process driving end (the first end of the drum frame body 11) 11/1, and a process non-driving end (the second end of the drum frame body 11) 11/2 opposite to the process driving end 11/1. The process driving end 11/1 and the process non-driving end 11/2 are the portions (ends) situated at the outermost positions in the drum frame body 11 in the rotation axis direction of the photosensitive drum 12. A plurality of the process driving ends 11/1 and a plurality of the process non-driving ends 11/2 may be present. The center of the drum frame body 11 in the rotation axis direction of the photosensitive drum 12 is referred to as a center 11/3. The distance between the center 11/3 and the process driving end 11/1 of the drum frame body 11 and the distance between the center 11/3 and the process non-driving end 11/2 thereof are equal to each other.

[0088] In the present Example, the process driving end 11/1 and the process non-driving end 11/2 are portions (ends) situated at the outermost positions of the process cartridge P in the rotation axis direction of the photosensitive drum 12. In other words, in the rotation axis direction of the photosensitive drum 12, the process driving end 11/1 and the process non-driving end 11/2 are in agreement with the driving end (the first end of the process cartridge P) and the non-driving end (the second end of the process cartridge P) of the process cartridge P. In the rotation axis direction of the photosensitive drum 12, the “driving side” of the drum frame body 11 or the process cartridge P is situated opposite to the “non-driving side” of the drum frame body 11 or the process cartridge P.

[0089] In the rotation axis direction of the photosensitive drum 12, the side on which the process driving end 11/1 is arranged with respect to the center 11/3 of the drum frame body 11 is the driving side of the drum frame body 11 or the driving side of the process cartridge P. In the rotation axis direction of the photosensitive drum 12, the side on which the process non-driving end 11/2 is arranged with respect to the center 11/3 of the drum frame body 11 is the non-driving side of the drum frame body 11 or the non-driving side of the process cartridge P. In the present Example, in the rotation axis direction of the photosensitive drum 12, the center 11/3 of the drum frame body 11 is the same as the center of the process cartridge P.

[0090] Although described later, the toner transport member 62 shown in FIG. 14 is rotatable around the rotation axis 62a. The direction of extension of the rotation axis 62a is referred to as the rotation axis direction (axis direction) of the toner transport member 62. Further, the toner transport screw 63 shown in FIG. 14 is rotatable around the rotation axis 63a. The direction of extension of the rotation axis 63a is referred to as the rotation axis direction (axis direction) of the toner transport screw 63.

[0091] As shown in FIG. 10, with regard to the rotation axis direction LD of the toner transport screw 63, the toner frame body 55 has a toner driving end (the first end of the toner frame body 55) 55a1, and a toner non-driving end 55a2 (the second end of the toner frame body 55) opposite to the toner driving end 55a1. The toner driving end 55a1 and the toner non-driving end 55a2 are portions (ends) situated at the outermost positions in the toner frame body 55 in the rotation axis direction of the toner transport screw 63. A plurality of the toner driving ends 55a1 and a plurality of the toner non-driving ends 55a2 may be present. In the rotation axis direction of the toner transport screw 63, the center of the toner frame body 55 is referred to as a center 55a3. The distance between the center 55a3 and the toner driving end 55a1 of the toner frame body 55 and the distance between the center 11/3 and the toner non-driving end 55a2 thereof are equal to each other.

[0092] In the present Example, the toner driving end 55a1 and the toner non-driving end 55a2 are portions (ends) situated at the outermost position in the toner cartridge T in the rotation axis direction of the toner transport screw 63. In other words, in the rotation axis direction of the toner transport screw 63, the toner driving end 55a1 and the toner non-driving end 55a2 are in agreement with the driving end (the first end of the toner cartridge T) and the non-driving end (the second end of the toner cartridge T) of the toner cartridge T.

[0093] In the rotation axis direction of the toner transport screw 63, the side on which the toner driving end 55a1 is arranged with respect to the center 55a3 of the toner frame body 55 is the driving side of the toner frame body 55, or the driving side of the toner cartridge T. In the rotation axis direction of the toner transport screw 63, the side on which the toner non-driving end 55a2 is arranged with respect to the center 55a3 of the toner frame body 55 is the non-driving side of the toner frame body 55, or the non-driving side of the toner cartridge T. In the present Example, in the rotation axis direction of the toner transport screw 63, the center 55a3 of the toner frame body 55 is the same as that of the toner cartridge T.

[0094] In the rotation axis direction of the toner transport screw 63, the driving side of the toner frame body 55 or the

driving side of the toner cartridge T is situated opposite to the non-driving side of the toner frame body 55 or the non-driving side of the toner cartridge T.

[0095] In the present Example, the rotation axis direction of the photosensitive drum 12, the rotation axis direction of the toner transport member 62, and the rotation axis direction of the toner transport screw 63 are in parallel to one another. Therefore, the rotation axis direction of the photosensitive drum 12, the rotation axis direction of the toner transport member 62, and the rotation axis direction of the toner transport screw 63 are each also referred to simply as the axis direction (the first direction) LD.

[0096] In the present Example, in the axis direction LD, the position of the center 55a3 of the toner frame body 55 and the position of the center 11/3 of the drum frame body 11 are the same. However, the position of the center 55a3 of the toner frame body 55 and the position of the center 11/3 of the drum frame body 11 may be different from each other.

[0097] As shown in FIGS. 3 and 5, the process cartridge P has a driving side process guide 22 on the driving side of the drum frame body 11. The toner cartridge T has a driving side toner guide 51 on the driving side of the toner frame body 55. As shown in FIGS. 4 and 6, the process cartridge P has a non-driving side process guide 23 on the non-driving side of the drum frame body 11. The toner cartridge T has a non-driving side toner guide 52 on the non-driving side of the toner frame body 55.

[0098] The direction in which the process cartridge P is attached with respect to the apparatus main body 100A is referred to as the attachment direction PDA. The direction in which the process cartridge P is removed from the apparatus main body 100A is referred to as the removal direction PDD. The attachment direction PDA and the removal direction PDD are collectively referred to as the removal/attachment direction PD of the process cartridge P. The driving side process guide 22 and the non-driving side process guide 23 are formed along the removal/attachment direction PD. The driving side process guide 22 and the non-driving side process guide 23 are guided by a guide portion of the apparatus main body 100A, and the process cartridge P moves in the removal/attachment direction PD with respect to the apparatus main body 100A.

[0099] The direction in which the toner cartridge T is attached to the apparatus main body 100A is referred to as an attachment direction TDA. The direction in which the toner cartridge T is removed from the apparatus main body 100A is referred to as a removal direction TDD. The attachment direction TDA and the removal direction TDD are collectively referred to as a removal/attachment direction TD of the toner cartridge T. The driving side toner guide 51 and the non-driving side toner guide 52 are formed along the removal/attachment direction TD. The driving side toner guide 51 and the non-driving side toner guide 52 are guided by the guide portion of the apparatus main body 100A, and the toner cartridge T moves in the removal/attachment direction TD with respect to the apparatus main body 100A.

[0100] In the present Example, the removal/attachment direction PD of the process cartridge P is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the removal/attachment direction PD is preferably smaller than the angle formed between the axis direction LD and the

removal/attachment direction PD, and the removal/attachment direction PD is more preferably the direction orthogonal to the axis direction LD.

[0101] In the present Example, the removal/attachment direction TD of the toner cartridge T is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the removal/attachment direction TD is preferably smaller than the angle formed between the axis direction LD and the removal/attachment direction TD, and the removal/attachment direction TD is more preferably the direction orthogonal to the axis direction LD.

[0102] Incidentally, in the present Example, although the removal/attachment direction PD and the removal/attachment direction TD are in parallel with each other, the removal/attachment direction PD and the removal/attachment direction TD may be different from each other.

[0103] In the present Example, the attachment and the removal of the process cartridge P are performed with the toner cartridge T not attached to the apparatus main body 100A. In other words, the attachment and the removal of the process cartridge P are performed before the attachment of the toner cartridge T with respect to the apparatus main body 100A.

[0104] With the toner cartridge T not attached with respect to the apparatus main body 100A, the process cartridge P is attached with respect to the apparatus main body 100A through the opening 100C. Further, with the process cartridge P attached with respect to the apparatus main body 100A, the toner cartridge T is attached with respect to the apparatus main body 100A and the process cartridge P through the opening 100C.

[0105] With the toner cartridge T and the process cartridge P attached with respect to the apparatus main body 100A, in the attachment direction PDA and the attachment direction TDA, the process cartridge P is situated on the downstream side of the toner cartridge T, and the toner cartridge T is situated on the upstream side of the process cartridge P.

[0106] When the toner cartridge T and the process cartridge P are removed from the apparatus main body 100A, the toner cartridge T is removed from the apparatus main body 100A and the process cartridge P through the opening 100C. Subsequently, the process cartridge P is removed from the apparatus main body 100A through the opening 100C.

Toner Cartridge

[0107] Referring to FIGS. 7, 8, and 11A and 11B to 17, a configuration of the toner cartridge T in accordance with the present Example will be described in further details.

[0108] FIGS. 11A, 11B, 12A, and 12B are each a perspective view of the toner cartridge T. Specifically, FIGS. 11A and 11B are each a perspective view of the toner cartridge T as seen from the driving side. In FIG. 11B, a part of the toner cartridge T is omitted. FIGS. 12A and 12B are each a perspective view of the toner cartridge T as seen from the non-driving side. In FIG. 12B, a part of the toner cartridge T is omitted.

[0109] FIG. 13 is a view of the toner cartridge T as seen in the removal direction TDD. FIG. 14 is a view showing an inside structure of the toner cartridge T. FIG. 15 is a cross sectional view of the toner cartridge T with the direction orthogonal to the axis direction LD as the cross sectional direction.

[0110] FIGS. 16A, 16B, and 17 are each a side view of the toner cartridge T. Specifically, FIGS. 16A and 16B are each a side view of the driving side of the toner cartridge T as seen in the axis direction LD. FIG. 17 is a side view of the non-driving side of the toner cartridge T as seen in the axis direction LD.

[0111] As shown in FIGS. 7 and 14, the toner frame body 55 of the toner cartridge T is provided with a toner accommodation chamber 53 (a first chamber) and a toner collecting chamber 54 (a second chamber). The toner to be supplied to the process cartridge P is accommodated in the toner accommodation chamber 53. The toner returned from the process cartridge P by a return screw 18 shown in FIG. 8 is accommodated in the toner collecting chamber 54.

[0112] The toner collecting chamber 54 and the toner accommodation chamber 53 are separated (isolated) from each other. Specifically, the toner frame body 55 has a partition wall 55b, and the toner collecting chamber 54 and the toner accommodation chamber 53 are completely partitioned by the partition wall 55b. This prevents the toner from moving between the toner collecting chamber 54 and the toner accommodation chamber 53, so that the toner accommodated in the toner collecting chamber 54 and the toner accommodated in the toner accommodation chamber 53 are prevented from being mixed with each other.

[0113] The volume of the toner accommodation chamber 53 is larger than the volume of the toner collecting chamber 54. In the present Example, in the axis direction LD, the distance between the partition wall 55b and the toner non-driving end 55a2 is shorter than the distance between the partition wall 55b and the toner driving end 55a1.

[0114] In the axis direction, the toner accommodation chamber 53 is arranged so as to overlap the center 55a3 of the toner frame body 55. On the other hand, in the axis direction LD, the toner collecting chamber 54 is arranged on the non-driving side of the toner frame body 55. In the axis direction LD, the distance between the toner collecting chamber 54 and the toner non-driving end 55a2 is shorter than the distance between the toner collecting chamber 54 and the toner driving end 55a1. Further, in the axis direction LD, the toner collecting chamber 54 is arranged between the toner accommodation chamber 53 and the toner non-driving end 55a2.

[0115] As shown in FIGS. 13 and 14, the toner frame body 55 has a container portion 56 including the toner accommodation chamber 53 and the toner collecting chamber 54, a driving end cover 57 arranged on the driving side of the toner cartridge T, and a non-driving end cover 58 arranged on the non-driving side of the toner cartridge T.

[0116] The container portion 56 includes a first container 56a provided with the toner accommodation chamber 53, and a second container 56b provided with the toner collecting chamber 54. In the present Example in which the second container 56b is fixed to the first container 56a, the sidewall of the first container 56a and the sidewall of the second container 56b are opposed to each other, and each have a function as the partition wall 55b.

[0117] As shown in FIGS. 11A and 11B, the toner frame body 55 has a toner discharge port (a first opening) 61 for discharging the toner accommodated in the toner accommodation chamber 53 toward the development unit 30 of the process cartridge P. The toner discharge port 61 establishes a communication between the outside of the toner frame body 55 and the toner accommodation chamber 53, so that

the toner accommodated in the toner accommodation chamber 53 is discharged to the outside of the toner frame body 55 through the toner discharge port 61. The toner discharge port 61 is included in the first container 56a.

[0118] In the axis direction LD, the toner discharge port 61 is arranged closer to the driving side of the toner frame body 55. In the axis direction LD, the distance between the toner discharge port 61 and the toner driving end 55a1 is shorter than the distance between the toner discharge port 61 and the toner non-driving end 55a2.

[0119] The toner discharge port 61 is covered with a discharge port shutter not shown. The discharge port shutter is opened/closed in conjunction with the attachment and the removal of the toner cartridge T with respect to the process cartridge P. With the toner cartridge T attached to the process cartridge P, the toner is allowed to be discharged from the toner discharge port 61. As described above, the development roller 32 of the process cartridge P develops the electrostatic latent image formed on the photosensitive drum 12 by the toner discharged from the toner discharge port 61.

[0120] As shown in FIGS. 7 and 14, the toner transport member 62 and the toner transport screw 63 are accommodated in the inside of the toner accommodation chamber 53. The toner transport member 62 and the toner transport screw 63 can be referred to as a first rotation member of the toner cartridge T.

[0121] As shown in FIGS. 14 and 15, the toner cartridge T has a toner discharge device 64 for discharging a toner from the toner discharge port 61. The toner discharge device 64 has a pump portion 65 for feeding air by compression, an air guide 66 for guiding the air fed by the pump portion 65 toward the toner discharge port 61, a compression portion 67 for compressing the pump portion 65, and an operation gear 68 for moving the compression portion 67. The operation gear 68 includes an operation groove 68a for moving the compression portion 67.

[0122] The toner transport member 62 has a fin 62b having flexibility, and a shaft 62c to which the fin 62b is attached, and rotates around the rotation axis 62a, thereby transporting a toner toward the toner transport screw 63. As shown in FIG. 15, the fin 62b comes in contact with the inner wall of the toner accommodation chamber 53, to be deformed like a broken line from the natural state indicated with a solid line, for transporting the toner. The toner transport screw 63 has a spiral fin 63b1 and a screw shaft 63b2, and rotates around the rotation axis 63a, thereby transporting the toner toward the toner discharge port 61. The spiral fin 63b1 and the screw shaft 63b2 are integrally formed.

[0123] On the other hand, the rotation of the operation gear 68 moves the compression portion 67 in engagement with the operation groove 68a in the direction of the rotation axis of the operation gear 68. The movement of the compression portion 67 compresses the pump portion 65, so that the pump portion 65 feeds air toward the toner accommodation chamber 53.

[0124] The fed air is guided toward the toner discharge port 61 by an air guide 66. Incidentally, a cover member is arranged in the vicinity of the toner discharge port 61, and the cover member covers a part of the toner transport screw 63. As a result of this, the toner transported by the toner transport member 62 is suppressed from being discharged directly from the toner discharge port 61, and the air released from the air guide 66 goes toward the toner discharge port 61 with efficiency.

[0125] Further, as shown in FIGS. 11A and 11B, the toner frame body 55 includes a toner receiving port (second opening) 84 for receiving the toner returned from the process cartridge P. The toner receiving port 84 establishes a communication between the outside of the toner frame body 55 and the toner collecting chamber 54. The toner collecting chamber 54 accommodates the toner (collected toner) discharged from a toner return opening 20 shown in FIG. 8, and having passed through the toner receiving port 84.

[0126] In the axis direction LD, the toner receiving port 84 is arranged closer to the non-driving side of the toner frame body 55. Namely, in the axis direction LD, the distance between the toner receiving port 84 and the toner non-driving end 55a2 is shorter than the distance between the toner receiving port 84 and the toner driving end 55a1. Further, in the axis direction LD, the distance between the toner receiving port 84 and the toner non-driving end 55a2 is shorter than the distance between the toner receiving port 84 and the center 55a3 of the toner frame body 55.

[0127] The toner receiving port 84 is covered with the receiving port shutter 85. The receiving port shutter 85 is opened/closed in conjunction with the attachment and the removal of the toner cartridge T with respect to the process cartridge P.

[0128] More specifically, the receiving port shutter 85 is movable between the closing position at which the toner receiving port 84 is covered and the opening position at which the toner receiving port 84 is exposed. As shown in FIG. 8, the attachment of the toner cartridge T to the process cartridge P brings the receiving port shutter 85 into contact with the process cartridge P, and causes the receiving port shutter 85 to move to the opening position. With the toner cartridge T attached to the process cartridge P, the toner receiving port 84 and the toner return opening 20 are opposed to each other.

[0129] As shown in FIG. 14, the toner collecting chamber 54 includes a first collecting chamber 69, a second collecting chamber 70, and a third collecting chamber 71. The toner cartridge T has a first collecting screw 72 for transporting a toner from the first collecting chamber 69 to the second collecting chamber 70, and a second collecting screw (second rotation member) 73 for transporting a toner from the second collecting chamber 70 to the third collecting chamber 71. The first collecting screw 72 and the second collecting screw 73 are accommodated in the toner collecting chamber 54.

[0130] As shown in FIG. 8, the first collecting screw 72 is rotatable around the rotation axis 72a, and the second collecting screw 73 is rotatable around the rotation axis 73a. The direction of the rotation axis 72a is in parallel with the axis direction LD, and the direction of the rotation axis 73a is the direction crossing the axis direction LD. The angle formed between the direction orthogonal to the axis direction LD and the direction of the rotation axis 73a is preferably smaller than the angle formed between the axis direction LD and the direction of the rotation axis 73a, and the direction of the rotation axis 73a is more preferably the direction orthogonal to the axis direction LD.

[0131] The toner received from the toner receiving port 84 is accumulated in the first collecting chamber 69, and is transported from the first collecting chamber 69 to the second collecting chamber 70 by the first collecting screw 72. The toner transported to the second collecting chamber

70 is transported upward in the vertical direction by the second collecting screw 73, and is transported from the second collecting chamber 70 to the third collecting chamber 71.

[0132] As shown in FIGS. 16A and 16B, the toner cartridge T has a second toner coupling 74, a first toner coupling 75, a transport gear 76, and a discharge gear 77.

[0133] The operation gear 68, the second toner coupling 74, the first toner coupling 75, the transport gear 76, and the discharge gear 77 are arranged on the driving side of the toner cartridge T. In other words, in the axis direction LD, the distance between the second toner coupling 74 or the first toner coupling 75, and the toner driving end 55a1 is shorter than the distance between the second toner coupling 74 or the first toner coupling 75, and the toner non-driving end 55a2.

[0134] The driving end cover 57 shown in FIGS. 13 and 16A covers at least a part of the operation gear 68, the second toner coupling 74, the first toner coupling 75, the transport gear 76, and the discharge gear 77, and has a driving side toner guide 51. The driving end cover 57 covers at least a part of the pump portion 65.

[0135] The transport gear 76 is coupled with the toner transport member 62, and the discharge gear 77 is coupled with the toner transport screw 63. The second toner coupling 74 receives a driving force (external force) from the apparatus main body 100A, and rotates the transport gear 76. As a result, the toner transport member 62 is rotated. The first toner coupling 75 is in engagement with the operation gear 68 via an idler gear, and the operation gear 68 is in engagement with the discharge gear 77. The first toner coupling 75 receives a driving force (external force) from the apparatus main body 100A, and rotates the operation gear 68 and the discharge gear 77. As a result, the pump portion 65 is compressed, so that the toner transport screw 63 is rotated.

[0136] In other words, the second toner coupling 74 has a function as a drive member for driving the toner transport member 62. The first toner coupling 75 has a function as a drive member for driving the toner discharge device 64 and the toner transport screw 63.

[0137] As shown in FIG. 17, the toner cartridge T has a detection member 78 for detecting the toner amount in the toner collecting chamber 54. The non-driving end cover 58 covers the detection member 78, and has the non-driving side toner guide 52.

[0138] The detection member 78 in the present Example is a light guide pair for guiding a light. The detection member 78 includes a light guide member 78a1 and a light guide member 78a2, and is arranged on the non-driving side of the toner cartridge T. In other words, in the axis direction LD, the distance between the light guide member 78a1 or the light guide member 78a2, and the toner non-driving end 55a2 is shorter than the distance between the light guide member 78a1 or the light guide member 78a2, and the toner driving end 55a1.

[0139] A part of the light guide member 78a1 and a part of the light guide member 78a2 are exposed at the toner collecting chamber 54. One of the light guide member 78a1 and the light guide member 78a2 guides a light toward the inside of the toner collecting chamber 54 from the outside of the toner collecting chamber 54. The light guided to the inside of the toner collecting chamber 54 passes through the toner collecting chamber 54. The other of the light guide

member **78a1** and the light guide member **78a2** guides a light to the outside of the toner collecting chamber **54** from the inside of the toner collecting chamber **54**.

[0140] The control portion **107** of the apparatus main body **100A** can detect the toner amount on the basis of the light that has passed through the inside of the toner collecting chamber **54** via the light guide member **78a1** and the light guide member **78a2**. Incidentally, as the detection member **78**, a mutually opposing electrode pair can be used. In this case, the control portion **107** of the apparatus main body **100A** can detect the toner amount on the basis of the change in electrostatic capacity between the electrode pair.

Process Cartridge

[0141] Referring to FIGS. **3** to **10**, the configuration of the process cartridge **P** in accordance with the present Example will be described in further details.

[0142] FIGS. **7** and **8** are each a cross sectional view of the process cartridge **P** and the toner cartridge **T**. Specifically, FIGS. **7** and **8** are each a cross sectional view of the process cartridge **P** and the toner cartridge **T** with the direction orthogonal to the axis direction **LD** assumed as the cross sectional direction. FIG. **8** is a view obtained by cutting the process cartridge **P** and the toner cartridge **T** along the rotation axis **RS** of a return screw **18** described later.

[0143] FIG. **9** is a side view for illustrating the driving transmission path of the process cartridge **P**. FIG. **10** is a view of the process cartridge **P** and the toner cartridge **T** as seen in the removal direction **PDD**.

[0144] The process cartridge **P** includes the development unit **30** and the drum unit **10**. The development unit **30** is movably (rotatably) coupled with respect to the drum unit **10**. As shown in FIGS. **5** and **6**, the process cartridge **P** has a driving side spring **37** and a non-driving side spring **38**, and the driving side spring **37** and the non-driving side spring **38** are attached to the drum unit **10** and the development unit **30**. The driving side spring **37** and the non-driving side spring **38** bias the development unit **30** so that the development roller **32** may be pressed against the photosensitive drum **12**.

[0145] As shown in FIGS. **7** and **8**, the development unit **30** has a development frame body **31**, a development roller **32** (a developer bearing member) for bearing a toner, a supply roller **33** (a supply member) for coming in contact with the development roller **32**, and supplying a toner, a development blade **34**, and a stirring unit **35** (a stirring member). The development frame body **31** supports the development roller **32**, the supply roller **33**, the development blade **34**, and the stirring unit **35**. The development frame body **31** is provided with a developer accommodation chamber **31a** and a development chamber **31b**.

[0146] The developer accommodation chamber **31a** is set at the top in the gravity direction of the development chamber **31b** via a wall surface portion **201e** that is a partition sheet shown in FIG. **1A**. Further, the wall surface portion **201e** is provided with an opening **201** for establishing a communication between the developer accommodation chamber **31a** and the development chamber **31b**.

[0147] In the developer accommodation chamber **31a**, a stirring unit **35** is arranged. In the development chamber **31b**, the development roller **32**, the supply roller **33**, and the development blade **34** are arranged.

[0148] The stirring unit **35** includes a rotatively supported stirring shaft **35c**, and a flexible sheet member **35a** fixed to the stirring shaft **35c**.

[0149] As shown in FIG. **7**, the supply roller **33** rotates in the same direction as that of the development roller **32** (an arrow **K** direction), and the stirring unit **35** rotates in the opposite direction to that of the supply roller **33** (an arrow **Q** direction). Further, the longitudinal direction of the supply roller **33** in the development unit **30** will be hereinafter referred to as a longitudinal direction.

[0150] The toner supplied from the toner cartridge **T** is accommodated in the developer accommodation chamber **31a**. The stirring unit **35** transports the toner accommodated in the developer accommodation chamber **31a** to the development chamber **31b**. The toner transported to the development chamber **31b** is supplied to the development roller **32** by the supply roller **33** that comes in contact with the development roller **32** to be rotated. The toner supplied to the development roller **32** is regulated by the development blade **34**, so that a toner layer is formed on the surface of the development roller **32**. The development blade **34** has a function as a layer thickness regulating member for regulating the thickness of the toner layer.

[0151] As shown in FIGS. **7** and **8**, the drum unit **10** has a drum frame body **11**, a photosensitive drum **12** (image bearing member), a charging roller **13**, a cleaning blade **14**, an intermediate transport member **15**, an intermediate screw **16**, a transmission shaft **17**, and a return screw **18**. The drum frame body **11** supports the photosensitive drum **12**, the charging roller **13**, the cleaning blade **14**, the intermediate transport member **15**, the intermediate screw **16**, the transmission shaft **17**, and the return screw **18**. Further, the drum unit **10** has a memory tag **90P** described later.

[0152] The drum frame body **11** includes a cleaning collecting chamber **19**. In the cleaning collecting chamber **19**, the intermediate transport member **15**, the intermediate screw **16**, and the return screw **18** are arranged.

[0153] As shown in FIG. **8**, the drum frame body **11** includes a return path **45** for accommodating the return screw **18**. The return path **45** can be said to be a part of the cleaning collecting chamber **19**.

[0154] The charging roller **13** is in contact with the photosensitive drum **12**, and is rotated by the photosensitive drum **12**. The cleaning blade **14** is in contact with the photosensitive drum **12**, and collects the toner left on the surface of the photosensitive drum **12**. The collected toner (the waste toner, the residual toner, or the recovered toner) is accommodated in the cleaning collecting chamber **19**. The collected toner is transported toward the intermediate screw **16** by the intermediate transport member **15**, and the intermediate screw **16** transports the collected toner toward the return screw **18**. The intermediate transport member **15** transports the collected toner toward the direction crossing the axis direction **LD**. The intermediate screw **16** transports the collected toner along the axis direction **LD**.

[0155] The return screw (rotation member) **18** rotates around the rotation axis (the second axis) **RS** shown in FIG. **6**. The direction of extension of the rotation axis **RS** of the return screw **18** is referred to as the rotation axis direction (the second direction) of the return screw **18**.

[0156] The rotation axis direction of the return screw **18** is the direction crossing the axis direction **LD**. The angle formed between the direction orthogonal to the axis direction **LD** and the rotation axis direction of the return screw **18**

is preferably smaller than the angle formed between the axis direction LD and the rotation axis direction of the return screw 18, and the rotation axis direction of the return screw 18 is more preferably the direction orthogonal to the axis direction LD.

[0157] As shown in FIG. 8, the drum frame body 11 includes a return opening 20. The return opening 20 establishes a communication between the return path 45 of the cleaning collecting chamber 19 and the outside of the drum frame body 11, and is opposed to the return screw 18. The collected toner delivered to the return screw 18 from the intermediate screw 16 is transported toward the return opening 20 by the return screw 18 and is discharged from the return opening 20, to pass through a toner receiving port 84 described later, and to be received by the toner cartridge T.

[0158] Thus, the return screw 18 has a function as a transport member for transporting the toner collected from the photosensitive drum 12 toward the toner receiving port 84. The direction of transport of the collected toner by the return screw 18 is the direction toward the toner cartridge T from the process cartridge P, and the upward direction in the vertical direction.

[0159] The return screw 18 has the spiral fin 18b and the screw shaft 18c, and rotates around the rotation axis RS, thereby transporting the toner toward the return opening 20. The spiral fin 18b and the screw shaft 18c are integrally formed.

[0160] As shown in FIGS. 3 and 5, the process cartridge P has a process coupling 36 (a first input portion or a development drive member), and a drum gear 21 (a second input portion or a drum drive member). The engagement of the process coupling 36 with the main body coupling of the apparatus main body 100A results in the transmission of a driving force (an external force) from the apparatus main body 100A to the process coupling 36. The engagement of the drum gear 21 with the main body gear of the apparatus main body 100A results in the transmission of a driving force (an external force) from the apparatus main body 100A to the drum gear 21, so that the drum gear 21 rotates. The rotation of the drum gear 21 drives and rotates the photosensitive drum 12.

[0161] In the present Example, the process coupling 36 and the drum gear 21 are arranged on the driving side of the process cartridge P. Namely, in the axis direction LD, the distance between the process driving end 11/1 and the process coupling 36 is shorter than the distance between the process non-driving end 11/2 and the process coupling 36. Similarly, in the axis direction LD, the distance between the process driving end 11/1 and the drum gear 21 is shorter than the distance between the process non-driving end 11/2 and the drum gear 21.

[0162] As shown in FIG. 9, the process cartridge P has a stirring gear 39 for driving the stirring unit 35, a development gear 40 for driving the development roller 32, and a supply gear 41 for driving the supply roller 33. The stirring gear 39, the development gear 40, and the supply gear 41 are coupled to the process coupling 36 via a plurality of the idler gears 42. The rotation of the process coupling 36 drives and rotates the development roller 32, the supply roller 33, and the stirring unit 35.

[0163] Further, the process cartridge P has an intermediate transport gear 24 for driving the intermediate transport member 15, an intermediate screw gear 25 for driving the intermediate screw 16, and a shaft gear 26 for driving the

transmission shaft 17. The intermediate transport gear 24, the intermediate screw gear 25, and the shaft gear 26 are coupled to the process coupling 36 via a plurality of the idler gears 27. The rotation of the process coupling 36 rotates the intermediate transport member 15, the intermediate screw 16, and the transmission shaft 17.

[0164] As shown in FIG. 8, the return path 45 and the return screw 18 are arranged on the non-driving side of the process cartridge P. In other words, in the axis direction LD, the distance between the return screw 18 and the process non-driving end 11/2 is shorter than the distance between the return screw 18 and the process driving end 11/1.

[0165] More specifically, as shown in FIG. 10, in the axis direction LD, the distance between the rotation axis RS of the return screw 18 and the process non-driving end 11/2 is shorter than the distance between the rotation axis RS and the process driving end 11/1. Whereas, in the axis direction LD, the distance between the rotation axis RS and the process non-driving end 11/2 is shorter than the distance between the rotation axis RS and the center 11/3 of the drum frame body 11.

[0166] As shown in FIG. 8, the transmission shaft 17 is attached to the transmission gear 28, and a return gear 29 to be in engagement with the transmission gear 28 is attached to the return screw 18. The transmission gear 28 and the return gear 29 are bevel gears, so that the rotation of the transmission shaft 17 rotates the return screw 18. Namely, the driving force transmitted to the process coupling 36 is transmitted from the driving side of the process cartridge P to the non-driving side of the process cartridge P by the transmission shaft 17, and is transmitted to the return screw 18. In other words, the process coupling 36 is configured so as to drive the return screw 18.

[0167] As shown in FIG. 10, the transmission shaft 17 is arranged outside the cleaning collecting chamber 19, and the transmission gear 28 and the return gear 29 are in engagement with each other outside the cleaning collecting chamber 19.

[0168] As shown in FIGS. 4 and 6, a development roller electrode 32a (a development roller contact), a development blade electrode 34a (a development blade contact), a supply roller electrode 33a (a supply roller contact) and a charging roller electrode 13a (a charging contact) are arranged on the non-driving side of the process cartridge P. In other words, the distance between the process non-driving end 11/2 and the charging roller electrode 13a is shorter than the distance between the process driving end 11/1 and the charging roller electrode 13a. Similarly, the distance between the process non-driving end 11/2, and the development roller electrode 32a, the development blade electrode 34a, or the supply roller electrode 33a is shorter than the distance between the process driving end 11/1, and the development roller electrode 32a, the development blade electrode 34a, or the supply roller electrode 33a.

[0169] The development roller electrode 32a, the development blade electrode 34a, the supply roller electrode 33a, and the charging roller electrode 13a are electrically connected with the development roller 32, the development blade 34, the supply roller 33, and the charging roller 13, respectively. When an image formation operation is performed, the development roller electrode 32a, the development blade electrode 34a, the supply roller electrode 33a,

and the charging roller electrode **13a** are applied with a prescribed voltage from the power supply portion **170** of the apparatus main body **100A**.

[0170] Each material for the development roller electrode **32a**, the development blade electrode **34a**, the supply roller electrode **33a**, and the charging roller electrode **13a** may be a metal, or may be a resin having conductivity.

[0171] Incidentally, in the present Example, the direction of each rotation axis of the development roller **32**, the supply roller **33**, the stirring unit **35**, the charging roller **13**, the intermediate transport member **15**, the intermediate screw **16**, and the transmission shaft **17** is in parallel with the axis direction LD. The direction of the rotation axis of each gear except for the process coupling **36**, the drum gear **21**, and the return gear **29** is also in parallel with the axis direction LD.

With Regard to Detailed Configuration of Inside of Development Unit **30** Characteristic of the Present Invention

[0172] Referring to FIGS. **1A** and **1B**, a description will be given to the shape of the opening **201** of the development unit **30** in accordance with the present invention. FIG. **1A** is a side surface cross sectional view of the development unit showing the shape of the opening **201** at the end in the longitudinal direction. FIG. **1B** is a side surface cross sectional view of the development unit showing the shape of the opening **201** at the central portion in the longitudinal direction.

[0173] As shown in FIG. **1A**, in the cross sectional view of the end in the longitudinal direction, the opening **201** is formed by being partitioned by a second end **201b** situated on the upstream side in a rotation direction Q of the stirring unit **35**, and a fourth end **201d** situated on the downstream side in the rotation direction Q of the stirring unit **35**. The second end **201b** is the opening end on the upstream side at the end in the longitudinal direction, and the fourth end **201d** is the opening end at the end on the downstream side in the longitudinal direction.

[0174] As shown in FIG. **1B**, in the cross sectional view of the central portion in the longitudinal direction, the opening **201** is formed by being partitioned by a first end **201a** situated on the upstream side in the rotation direction Q of the stirring unit **35**, and a third end **201c** situated on the downstream side in the rotation direction Q of the stirring unit **35**. The first end **201a** is the opening end on the upstream side at the central portion in the longitudinal direction, and the third end **201c** is the opening end on the downstream side at the central portion in the longitudinal direction.

[0175] The length of the sheet member **35a** of the stirring unit **35** will be described. The distance between the rotation axis center of the stirring unit **35** and the free tip portion of the sheet when the sheet member **35a** has no flexure is referred to as the “free distance of sheet”. Whereas, the distance between the rotation center of the stirring unit **35** and the first end **201a** is referred to as the “center-end distance at central portion”. Further, the distance between the rotation center of the stirring unit **35** and the second end **201b** is referred to as the “center-end distance at end”. At this step, the “free distance of sheet” is longer than the “center-end distance at central portion” and the “center-end distance at end”. Therefore, the sheet member **35a** of the stirring unit **35** is configured to come in contact with the wall surface portion **201e** in the vicinity of the opening **201** of the

developer accommodation chamber **31a** while bending, and to feed a developer of the developer accommodation chamber **31a** into the development chamber **31b** with stirring.

[0176] Herein, the contact region of the supply roller **33** with the development roller **32** is referred to as a contact region **202**. As shown in the cross sectional view, when the supply roller **33** bites into the development roller **32** by a prescribed distance, the contact region **202** may be at the position of the center of the biting region, may be the end, or may be at other prescribed positions. At this step, the position of the first end **201a** of the opening **201** is set on the side closer to the contact region **202** of the supply roller **33** than the second end **201b** of the opening **201**. Specifically, the relationship of “distance L1 between first end **201a** and contact region **202**” < “distance L2 between second end **201b** and contact region **202**” holds. At this step, the distance L1 and the distance L2 are not limited to the distance between the end and the contact region **202**, but may be the distance between the end and the inlet portion **202a**, or the distance between the end and an outlet portion **202b**.

[0177] Further, in FIG. **1A**, for measuring the distance L2, the distance between the line passing through the second end **201b**, and perpendicular to the wall surface portion **201e**, and the line passing through the contact region **202**, and perpendicular to the wall surface portion **201e** is measured. Whereas, in FIG. **1B**, for measuring the distance L1, the distance between the line passing through the first end **201a**, and perpendicular to the wall surface portion **201e**, and the line passing through the contact region **202**, and perpendicular to the wall surface portion **201e** is measured. However for example, not the line perpendicular to the wall surface portion **201e**, but the line connecting the second end **201b** (or the first end **201a**) and the rotation axis center of the stirring unit **35** may be used. Alternatively, the distance between the second end **201b** (or the first end **201a**) and the contact region **202** may be directly measured.

Additional Configuration

[0178] In addition to the configurations up to this point, further, in the present Example, each of the first end **201a** and the second end **201b** is set at a more effective position. Below, a description will be given by reference to the side surface cross sectional view of each development unit shown in FIGS. **18A** and **18B**. FIG. **18A** shows the shape of the opening **201** at each opposite end in the longitudinal direction, and FIG. **18B** shows the shape of the opening **201** at the central portion in the longitudinal direction.

[0179] As shown in FIGS. **18A** and **18B**, the line connecting the stirring shaft center and the supply roller center is referred to as L3. As shown in FIG. **18B**, the first end **201a** of the opening **201** is situated downstream of L3 in the rotation direction Q of the stirring unit **35**. On the other hand, as shown in FIG. **18A**, the second end **201b** of the opening **201** is situated upstream of L3 in the rotation direction Q of the stirring unit **35**.

[0180] Further, the size of the opening **201** in the present Example will be described by reference to FIG. **20**. FIG. **20** is a view of the opening **201** as seen in the front direction from the inside of the developer accommodation chamber **31a**. In the present Example, as shown in FIG. **20**, the opening **201** satisfies the relationship of “distance L4 between first end **201a** and third end **201c**” < “distance L5 between second end **201b** and fourth end **201d**”.

[0181] Herein, the shape of the opening 201 may be the shape as shown in FIGS. 22A and 22B. FIGS. 22A and 22B are each a view of the opening 201 as seen in the front direction from the inside of the developer accommodation chamber 31a, showing another shape example of the opening 201.

[0182] The shape without a partition of the opening 201 in the longitudinal direction as shown in FIG. 22A, or the shape in which the first end 201a and the second end 201b of the opening 201 continuously change in the longitudinal direction as shown in FIG. 22B is also acceptable. Further, the shape of the opening 201 is not limited to the examples so long as it satisfies the positional relationship previously described.

Description of Circulation of Developer Inside Development Unit 30

[0183] Below, referring to FIGS. 19A and 19B, a description will be given to the circulation of the developer inside the development unit 30. FIG. 19A is a cross sectional view of the development unit 30 as seen in the longitudinal direction. FIG. 19B is a cross sectional view at the line along the longitudinal direction, and shows the movement of the developer in the longitudinal direction in the development chamber 31b in the periphery of the supply roller 33 of the development unit 30.

[0184] As shown in FIG. 19A, the developer to be accommodated in the developer accommodation chamber 31a has its own weight applied in an arrow Z direction, and hence it becomes more compact as it moves toward the development chamber 31b. Further, the sheet member 35a of the stirring unit 35 comes in contact with the wall surface portion 201e of the opening 201, and rotates in an arrow Q direction while bending. As a result, the developer in the developer accommodation chamber 31a is fed in an arrow R direction with stirring inside the accommodation chamber. Particularly, at the central portion in the longitudinal direction, the amount of the developer to be fed from the developer accommodation chamber 31a to the development chamber 31b tends to be larger than at the end. Incidentally, in FIG. 19A, the tip (the end on the side not coupled with the shaft) of the sheet member 35a moves while being in contact with the opening 201 when passing through the vicinity of the opening 201. However, the length of the sheet member 35a may be shorter than this so long as the developer can be stirred. The length of the sheet member 35a may be just large enough that the tip does not come in contact with the opening 201 when it passes through the vicinity of the opening.

[0185] On the other hand, the developer in the development chamber 31b is included in the supply roller 33 from the inlet portion 202a. The developer included in the supply roller 33 is discharged from the outlet portion 202b, thereby circulating the developer in the development chamber 31b. The action of the supply roller 33 is referred to as the “breath of the supply roller 33”. Incidentally, in the present Example, in order to show the action of the supply roller 33 and the movement of the developer, a description has been given by distinguishing the contact region 202, the inlet portion 202a, and the outlet portion 202b. However, in actuality, respective locations are not clearly separable. For this reason, for describing the positional relationship and the acting direction in the present invention, these are not supposed to be distinguished and limited.

[0186] Further, referring to FIG. 19B, a description will be given to the movement of the developer in the longitudinal direction of the supply roller in the development chamber 31b. Of the developers in the development chamber 31b, the developer that was not included in the supply roller 33 in the vicinity of the inlet portion 202a by the rotation of the supply roller 33 is fed to the longitudinal end of the process cartridge (in the arrow S direction).

[0187] As described up to this point, the developer is circulated, and generally, the circulation of the developer is more preferably performed smoothly. However, an increase in weight of the developer in the developer accommodation chamber 31a results in the compaction of the lower part in the gravity direction in the developer accommodation chamber 31a. As a result, the amount of the developer to be fed to the development chamber 31b increases. Particularly, at the central portion in the longitudinal direction, the effect tends to increase. Then, in the development chamber 31b, the previous developer fed from the developer accommodation chamber 31a compacts the vicinity of the outlet portion 202b with the developer. This makes the “breath of the supply roller” impossible. As a result, the amount of the developer accumulated in the vicinity of the inlet portion 202a to be fed to the longitudinal end of the process cartridge (in the arrow S direction) by the rotation of the supply roller 33 increases.

[0188] Then, at the end in the longitudinal direction of the development chamber 31b, the circulation of the developer is less. For this reason, the developer becomes compact. As a result, an image defective due to toner deterioration may be caused.

Effect of the Present Invention

[0189] In consideration of the circulation of the developer described above, as shown in FIGS. 1A and 1B, the satisfaction of the relationship of distance L1 < distance L2 can reduce the amount of the developer to be transported into the development chamber 31b from the developer accommodation chamber 31a at the central portion in the longitudinal direction. As a result, it is possible to make proper the amount of the developer staying in the vicinity of the inlet portion 202a of the supply roller 33 in the development chamber 31b. For this reason, it is possible to suppress the amount of the developer to be fed to each longitudinal end of the supply roller 33 in the development chamber 31b (the arrow S direction (see FIGS. 19A and 19B)) by the rotation of the supply roller 33.

[0190] Further, as shown in FIG. 18B, at the central portion in the longitudinal direction, the position of the first end 201a is set at downstream in the rotation direction of the line L3. With this configuration, at the central portion in the longitudinal direction, the direction in which the transport force F1 of the developer by the sheet member 35a of the stirring unit 35 acts can be shifted from the direction toward the outlet portion 202b. For this reason, it is possible to more effectively reduce the amount of the developer to be transported into the development chamber 31b from the developer accommodation chamber 31a by the transport force of the stirring unit 35 at the longitudinal central portion of the supply roller.

[0191] On the other hand, as shown in FIG. 18A, at the opposite ends in the longitudinal direction, the position of the second end 201b is set at upstream of the line L3 in the rotation direction. With this configuration, the direction in

which the transport force **F2** of the tip portion **35b** of the bent sheet member **35a** acts can be turned in the direction of the outlet portion **202b** of the supply roller **33**. Further, the tip portion **35b** of the sheet member **35a** can enter the inside of the development chamber **31b**. As a result, the developer can be effectively circulated in the development chamber **31b** at both ends in the longitudinal direction.

[0192] Further, as shown in FIG. 20, by satisfying the relationship of distance **L4** < distance **L5** for the size of the opening **201**, it is possible to reduce the amount of the developer to be transported into the development chamber **31b** from the developer accommodation chamber **31a** at the central portion in the longitudinal direction to which the amount of the developer to be fed tends to be large. For this reason, it is possible to make the circulation of the whole development unit proper.

[0193] Up to this point, by achieving the configuration of the present Example, it is possible to provide a development unit that can make proper the circulation balance of the developer in the development unit, and that does not undergo the leakage of the developer or an image detriment due to a circulation failure.

[0194] Incidentally, the present invention is not limited to the replenishment configuration including the process cartridge and the toner cartridge previously described. As shown in FIGS. 21A and 21B, the same effect is also expected for an integral type cartridge including the process cartridge and the toner cartridge integrally. FIGS. 21A and 21B are each a cross sectional view showing the shape of the opening **201** with an all-in-one cartridge configuration in which the developer accommodation chamber **31a** is filled with a large volume of a developer, including the same configuration as that of the present Example therein. Such an integral type configuration can also provide the effect of making the circulation of the developer proper.

[0195] As described up to this point, with the configuration described in the present specification, in the development unit configuration in which the developer accommodation chamber having a stirring unit is set at the top in the gravity direction of the development chamber, even when the weight of the developer increases, it is possible to keep the circulation of the developer proper. Namely, the opening coupling the developer accommodation chamber and the development chamber is in the shape that allows the developer to pass therethrough more easily at the ends in the longitudinal direction than at the central portion. Thus, by making the position and the shape of the opening different between at the ends in the longitudinal direction and at the central portion, it is possible to properly circulate the developer. Typically, for example, as described in FIGS. 20, 22A, and 22B, it is configured such that the frontage is wider at each end in the longitudinal direction than at the central portion. However, it is essential only that the shape of the opening is the shape that allows the developer to pass more easily at each end in the longitudinal direction than at the central portion. For example, it is only required that a part of the end has a portion with a larger frontage than that of the central portion.

[0196] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims

is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0197] This application claims the benefit of Japanese Patent Application No. 2024-033700, filed on Mar. 6, 2024, which is hereby incorporated by reference wherein in its entirety.

What is claimed is:

1. A developing apparatus comprising:

- a developer bearing member for bearing a developer;
- a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;
- a development chamber including the developer bearing member and the supply member arranged therein;
- a developer accommodation chamber for accommodating the developer; and
- a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft,

wherein,

the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus,

the developer bearing member, the supply member, and the stirring member are rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center,

a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and

in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

2. The developing apparatus according to claim 1, wherein

in a cross section in the longitudinal direction, the first end is situated on a downstream side in a rotation direction of the stirring member than a line connecting a rotation center of the shaft and a rotation center of the supply member, and

the second end is situated on an upstream side in the rotation direction of the stirring member than the line connecting a rotation center of the shaft and the rotation center of the supply member.

3. The developing apparatus according to claim 1, wherein

the sheet member of the stirring member has flexibility, a distance between a rotation center of the shaft and a free tip portion of the sheet member when the sheet member has no flexure is longer than a distance between the rotation center and the wall surface portion, and the sheet member rotates while coming in contact with the wall surface portion.

4. The developing apparatus according to claim 3, wherein

the sheet member can enter the opening at an end in the longitudinal direction, and does not enter the opening at a central portion in the longitudinal direction.

5. The developing apparatus according to claim 1, wherein

in a cross section in the longitudinal direction, a distance between the first end at a central portion in the longitudinal direction and a third end is shorter than a distance between the second end at an end in the longitudinal direction and a fourth end where the third end represents an end on the downstream side at a central portion in the longitudinal direction of the opening, and the fourth end represents an end on the downstream side at an end in the longitudinal direction.

6. An image forming apparatus comprising:

a developer bearing member for bearing a developer; a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;

a development chamber including the developer bearing member and the supply member arranged therein;

a developer accommodation chamber for accommodating the developer;

a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft; and

an image bearing member for developing an electrostatic latent image formed on a surface when the developer is supplied from the developer bearing member,

wherein,

the developer accommodation chamber is set at a top in a gravity direction of the development chamber,

the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction as a center,

a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and

in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a distance between a first end and a contact region of the developer bearing member and the supply member is shorter than a distance between a second end and the contact region where the first end represents the end on the upstream side at a central portion in the longitudinal direction, and the second end represents the end on the upstream side at the end in the longitudinal direction.

7. A developing apparatus comprising:

a developer bearing member for bearing a developer;

a supply member for coming in contact with the developer bearing member, thereby forming a contact region, and supplying the developer to a surface of the developer bearing member in the contact region;

a development chamber including the developer bearing member and the supply member arranged therein;

a developer accommodation chamber for accommodating the developer; and

a stirring member having a shaft, and a sheet member held by the shaft, the sheet member stirring the developer accommodated in the developer accommodation chamber by rotation of the shaft,

wherein,

the developer accommodation chamber is set at a top in a gravity direction of the development chamber in a posture in which the developing apparatus is mounted in an image forming apparatus,

the developer bearing member, the supply member, and the stirring member are each rotatable around a rotation axis along a longitudinal direction of the developing apparatus as a center,

a wall surface portion for partitioning the developer accommodation chamber and the development chamber is provided with an opening for enabling a movement of the developer from the developer accommodation chamber to the development chamber along the longitudinal direction, and

in a cross section in the longitudinal direction, the opening has an end on an upstream side and an end on a downstream side in a direction of passage of the sheet member according to rotation of the stirring member, and a width between the end on the upstream side and the end on the downstream side at a central portion in the longitudinal direction is narrower than a width between the end on the upstream side and the end on the downstream side at the ends in the longitudinal direction.

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